Linear Discriminant Analysis (LDA)

Nathan Seto

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Predicting Problematic Internet Use with LDA

The provided dataset, 'train.csv', contained a large amount of NA values in all variables.

A preview of the data (First 20 rows of columns 2 - 10):

2	3	4	5	6	7	8	9	10
Fall	5	0	Winter	51	Fall	16.87732	46.00	50.8
Summer	9	0	NA	NA	Fall	14.03559	48.00	46.0
Summer	10	1	Fall	71	Fall	16.64870	56.50	75.6
Winter	9	0	Fall	71	Summer	18.29235	56.00	81.6
Spring	18	1	Summer	NA	NA	NA	NA	NA
Spring	13	1	Winter	50	Summer	22.27995	59.50	112.2
Fall	10	0	NA	NA	Fall	19.66076	55.00	84.6
Fall	10	1	NA	NA	Fall	16.86129	59.25	84.2
Summer	15	0	NA	NA	Spring	NA	NA	NA
Summer	19	1	Summer	NA	NA	NA	NA	NA
Spring	11	1	NA	NA	NA	NA	NA	NA
Fall	11	0	Summer	66	NA	NA	NA	NA
Fall	13	0	NA	NA	Winter	21.07907	57.75	100.0
Spring	12	0	NA	NA	Spring	15.54411	60.00	79.6
Spring	12	0	NA	NA	NA	NA	NA	NA
Winter	7	0	Summer	51	Spring	29.31577	54.00	121.6
Spring	5	1	Summer	80	Spring	17.28450	44.00	47.6
Fall	10	1	NA	NA	Fall	19.89316	55.00	85.6
Winter	6	0	Spring	60	Winter	30.09465	37.50	60.2
Winter	10	0	NA	NA	NA	NA	NA	NA
Winter	14	0	Spring	68	Spring	17.16864	66.50	108.0
Summer	16	0	NA	NA	Summer	29.28586	71.00	210.0
Fall	12	0	Winter	58	Fall	34.18728	60.50	178.0
Winter	13	0	Summer	72	Winter	20.61306	69.00	139.6
Summer	15	0	Fall	70	Summer	21.17778	62.00	115.8

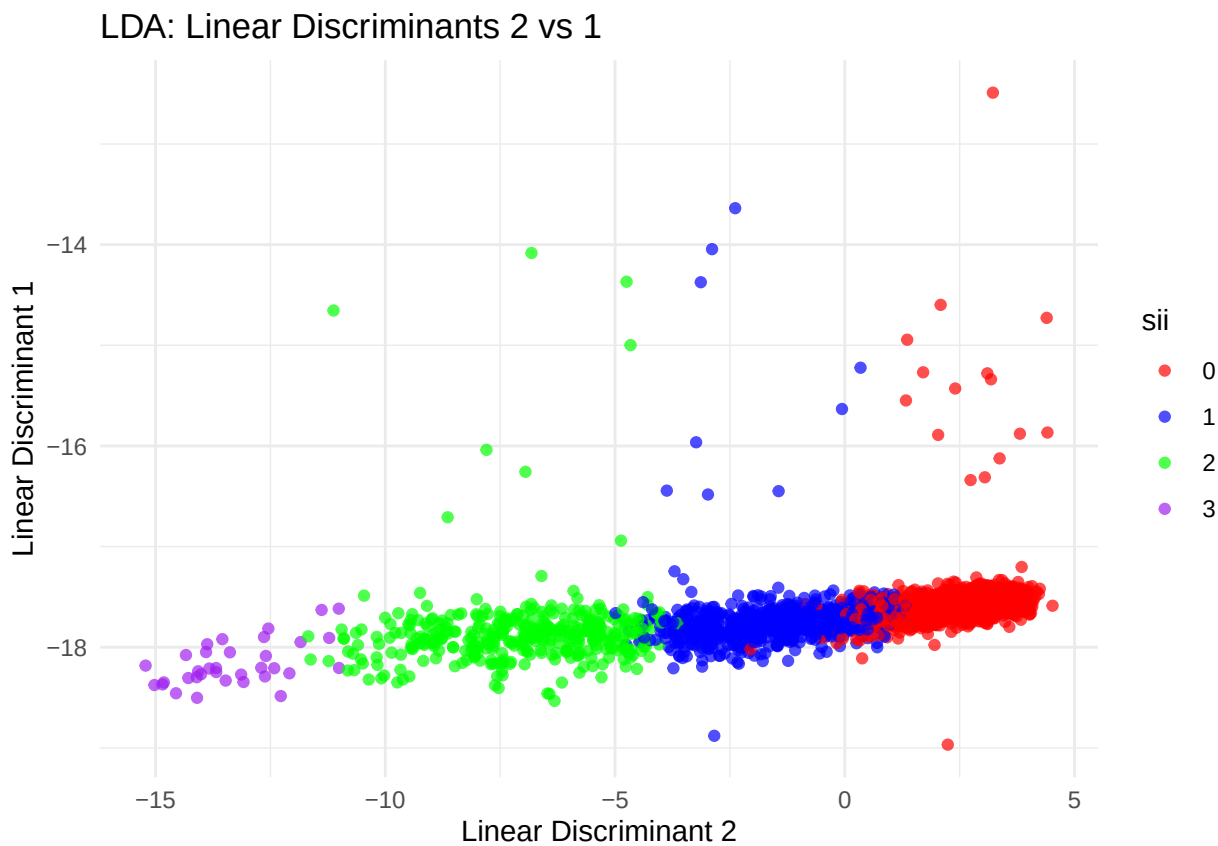
LDA with Full Imputation

The first method I used was to impute all the NA values in the dataset.

```
## [1] "Confusion Matrix on Train (Full Imputation)"
```

	0	1	2	3	Missing
0	1549	44	0	0	1
1	41	677	12	0	0
2	0	27	340	11	0
3	0	0	2	32	0
Missing	0	0	0	0	1224

```
## [1] "Accuracy (Train, Full Imputation): 96.52 %"
```




```
## [1] "Confusion Matrix (Test, Full Imputation):"
```

	0	1	2	3	Missing
0	6	0	0	0	0
1	0	3	0	0	0
2	0	0	2	0	0
3	0	0	0	0	0
Missing	0	0	0	0	9

```
## [1] "Accuracy (Test, Full Imputation): 100 %"
```

LDA when Removing NA from 'sii' Early

I noticed that in rows where `sii` was NA, many of the other cells in the row were also NA. So, here, I started by removing the rows where `sii` = NA. After doing so, R gave me an error that some predictors were colinear, so I removed the colinear columns before fitting the model.

```r
library(recipes)
library(readr)
library(dplyr)
library(caret)
library(ggplot2)
library(parsnip)
library(purrr)
library(discrim)
library(tidyr)
library(lda)
library(VIM)
library(MASS)
set.seed(445)
train = read_csv("train.csv")
```

```
## Rows: 3960 Columns: 82
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (12): id, Basic_Demos-Enroll_Season, CGAS-Season, Physical-Season, Fitne...
## dbl (70): Basic_Demos-Age, Basic_Demos-Sex, CGAS-CGAS_Score, Physical-BMI, P...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
train_clean = train[!is.na(train$sii), ]

colnames(train) = gsub("-", "_", colnames(train)) # Replacing hyphens because R can't handle those
train_clean = train_clean[, !grepl("Season", names(train))]
train_clean$sii = factor(train_clean$sii) # Making sure `sii` is a factor with 4 levels (plus extra lev

categorical_vars = train_clean[, sapply(train_clean, function(x) is.factor(x) | is.character(x))]
# Select categorical variables including 'sii'
# Identify columns that contain 'PCIAT' in their names but excluding 'PCIAT_PCIAT_Total' (quant. variab
pciat_columns = grep("PCIAT", names(train_clean), value = TRUE)
pciat_columns = pciat_columns[pciat_columns != "PCIAT_PCIAT_Total"]
categorical_vars = cbind(categorical_vars, train_clean[, pciat_columns])

categorical_vars = categorical_vars %>%
  mutate(across(everything(), ~ replace(., is.na(.), "Missing")))
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `across(everything(), ~replace(., is.na(.), "Missing"))`.
## Caused by warning in `[<-.factor`:
## ! invalid factor level, NA generated
```

```r
quantitative_vars = train_clean[, sapply(train_clean, is.numeric)]

quantitative_vars_imputed = quantitative_vars %>%
  mutate(across(where(is.numeric), ~ ifelse(is.na(.), median(., na.rm = TRUE), .)))

#train_data = bind_cols(categorical_vars, quantitative_vars_imputed)

# ---------------------------------------------------------------------------------------------------
# Removing collinear variables:
```

```r
# Compute correlation matrix for quantitative variables
correlation_matrix = cor(quantitative_vars_imputed)

# Find highly correlated pairs
high_corr_pairs = findCorrelation(correlation_matrix, cutoff = 0.8, verbose = TRUE)
```

```
## Compare row 66  and column  48 with corr  0.822
##   Means:  0.287 vs 0.141 so flagging column 66
## Compare row 5  and column  6 with corr  0.809
##   Means:  0.217 vs 0.137 so flagging column 5
## Compare row 6  and column  4 with corr  0.845
##   Means:  0.202 vs 0.135 so flagging column 6
## Compare row 63  and column  61 with corr  0.838
##   Means:  0.2 vs 0.133 so flagging column 63
## Compare row 41  and column  42 with corr  0.994
##   Means:  0.195 vs 0.131 so flagging column 41
## Compare row 42  and column  39 with corr  0.997
##   Means:  0.174 vs 0.129 so flagging column 42
## Compare row 39  and column  32 with corr  0.993
##   Means:  0.16 vs 0.128 so flagging column 39
## Compare row 32  and column  43 with corr  0.994
##   Means:  0.147 vs 0.127 so flagging column 32
## Compare row 43  and column  31 with corr  1
##   Means:  0.128 vs 0.127 so flagging column 43
## Compare row 31  and column  34 with corr  1
##   Means:  0.11 vs 0.127 so flagging column 34
## Compare row 31  and column  33 with corr  0.999
##   Means:  0.095 vs 0.128 so flagging column 33
## Compare row 31  and column  37 with corr  0.979
##   Means:  0.079 vs 0.129 so flagging column 37
## Compare row 31  and column  40 with corr  0.999
##   Means:  0.063 vs 0.131 so flagging column 40
## Compare row 31  and column  29 with corr  0.99
##   Means:  0.046 vs 0.134 so flagging column 29
## Compare row 18  and column  16 with corr  0.867
##   Means:  0.127 vs 0.137 so flagging column 16
## Compare row 22  and column  24 with corr  0.914
##   Means:  0.106 vs 0.137 so flagging column 24
## Compare row 68  and column  67 with corr  0.996
##   Means:  0.094 vs 0.139 so flagging column 67
## All correlations <= 0.8
```

```r
# Remove highly correlated variables
quantitative_vars_filtered = quantitative_vars_imputed[, -high_corr_pairs]

train_data = bind_cols(categorical_vars, quantitative_vars_filtered)
```

```
## New names:
## * `PCIAT-PCIAT_01` -> `PCIAT-PCIAT_01...3`
## * `PCIAT-PCIAT_02` -> `PCIAT-PCIAT_02...4`
## * `PCIAT-PCIAT_03` -> `PCIAT-PCIAT_03...5`
## * `PCIAT-PCIAT_04` -> `PCIAT-PCIAT_04...6`
## * `PCIAT-PCIAT_05` -> `PCIAT-PCIAT_05...7`
## * `PCIAT-PCIAT_06` -> `PCIAT-PCIAT_06...8`
## * `PCIAT-PCIAT_07` -> `PCIAT-PCIAT_07...9`
```

```
## * `PCIAT-PCIAT_08` -> `PCIAT-PCIAT_08...10`
## * `PCIAT-PCIAT_09` -> `PCIAT-PCIAT_09...11`
## * `PCIAT-PCIAT_10` -> `PCIAT-PCIAT_10...12`
## * `PCIAT-PCIAT_11` -> `PCIAT-PCIAT_11...13`
## * `PCIAT-PCIAT_12` -> `PCIAT-PCIAT_12...14`
## * `PCIAT-PCIAT_13` -> `PCIAT-PCIAT_13...15`
## * `PCIAT-PCIAT_14` -> `PCIAT-PCIAT_14...16`
## * `PCIAT-PCIAT_15` -> `PCIAT-PCIAT_15...17`
## * `PCIAT-PCIAT_16` -> `PCIAT-PCIAT_16...18`
## * `PCIAT-PCIAT_17` -> `PCIAT-PCIAT_17...19`
## * `PCIAT-PCIAT_19` -> `PCIAT-PCIAT_19...21`
## * `PCIAT-PCIAT_20` -> `PCIAT-PCIAT_20...22`
## * `PCIAT-PCIAT_01` -> `PCIAT-PCIAT_01...55`
## * `PCIAT-PCIAT_02` -> `PCIAT-PCIAT_02...56`
## * `PCIAT-PCIAT_03` -> `PCIAT-PCIAT_03...57`
## * `PCIAT-PCIAT_04` -> `PCIAT-PCIAT_04...58`
## * `PCIAT-PCIAT_05` -> `PCIAT-PCIAT_05...59`
## * `PCIAT-PCIAT_06` -> `PCIAT-PCIAT_06...60`
## * `PCIAT-PCIAT_07` -> `PCIAT-PCIAT_07...61`
## * `PCIAT-PCIAT_08` -> `PCIAT-PCIAT_08...62`
## * `PCIAT-PCIAT_09` -> `PCIAT-PCIAT_09...63`
## * `PCIAT-PCIAT_10` -> `PCIAT-PCIAT_10...64`
## * `PCIAT-PCIAT_11` -> `PCIAT-PCIAT_11...65`
## * `PCIAT-PCIAT_12` -> `PCIAT-PCIAT_12...66`
## * `PCIAT-PCIAT_13` -> `PCIAT-PCIAT_13...67`
## * `PCIAT-PCIAT_14` -> `PCIAT-PCIAT_14...68`
## * `PCIAT-PCIAT_15` -> `PCIAT-PCIAT_15...69`
## * `PCIAT-PCIAT_16` -> `PCIAT-PCIAT_16...70`
## * `PCIAT-PCIAT_17` -> `PCIAT-PCIAT_17...71`
## * `PCIAT-PCIAT_19` -> `PCIAT-PCIAT_19...72`
## * `PCIAT-PCIAT_20` -> `PCIAT-PCIAT_20...73`
```

```r
quantitative_vars_filtered = quantitative_vars_filtered %>% mutate(sii = train_data$sii)

# ----------------------------------------------------------------------------------------------------
# Linear Discriminant Analysis

lda_data_train = train_data
lda_data_train$sii = as.factor(lda_data_train$sii)  # Ensure `sii` is a factor for LDA

lda_data_train = subset(lda_data_train, select = -id)


# Perform Linear Discriminant Analysis
lda_model_train = lda(sii ~ ., data = lda_data_train)
```

```
## Warning in lda.default(x, grouping, ...): variables are collinear
```

```r
# View the LDA model output
print(lda_model_train)
```

```
## Call:
## lda(sii ~ ., data = lda_data_train)
##
## Prior probabilities of groups:
```

```
##         0         1         2         3
## 0.5826023 0.2668129 0.1381579 0.0124269
##
## Group means:
##   `PCIAT-PCIAT_01...3`1 `PCIAT-PCIAT_01...3`2 `PCIAT-PCIAT_01...3`3
## 0            0.18005019            0.28419072             0.1242158
## 1            0.06301370            0.17534247             0.2520548
## 2            0.01322751            0.04761905             0.0952381
## 3            0.02941176            0.00000000             0.0000000
##   `PCIAT-PCIAT_01...3`4 `PCIAT-PCIAT_01...3`5 `PCIAT-PCIAT_01...3`Missing
## 0            0.07465496            0.02132999                 0.001254705
## 1            0.29863014            0.16027397                 0.001369863
## 2            0.36772487            0.46031746                 0.000000000
## 3            0.17647059            0.79411765                 0.000000000
##   `PCIAT-PCIAT_02...4`1 `PCIAT-PCIAT_02...4`2 `PCIAT-PCIAT_02...4`3
## 0            0.18444166            0.26035132            0.09786700
## 1            0.05068493            0.20000000            0.29726027
## 2            0.01587302            0.02910053            0.09259259
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_02...4`4 `PCIAT-PCIAT_02...4`5 `PCIAT-PCIAT_02...4`Missing
## 0            0.04203262           0.008155583                 0.001254705
## 1            0.27671233           0.135616438                 0.000000000
## 2            0.41005291           0.436507937                 0.000000000
## 3            0.05882353           0.941176471                 0.000000000
##   `PCIAT-PCIAT_03...5`1 `PCIAT-PCIAT_03...5`2 `PCIAT-PCIAT_03...5`3
## 0            0.17377666            0.31555834             0.1336261
## 1            0.01506849            0.19452055             0.3438356
## 2            0.00000000            0.01851852             0.1031746
## 3            0.00000000            0.00000000             0.0000000
##   `PCIAT-PCIAT_03...5`4 `PCIAT-PCIAT_03...5`5 `PCIAT-PCIAT_03...5`Missing
## 0            0.05646173            0.01129235                 0.001254705
## 1            0.34383562            0.09726027                 0.002739726
## 2            0.46031746            0.41269841                 0.002645503
## 3            0.05882353            0.94117647                 0.000000000
##   `PCIAT-PCIAT_04...6`1 `PCIAT-PCIAT_04...6`2 `PCIAT-PCIAT_04...6`3
## 0            0.21329987            0.04328733           0.008155583
## 1            0.36712329            0.18767123           0.054794521
## 2            0.23809524            0.21957672           0.129629630
## 3            0.02941176            0.11764706           0.117647059
##   `PCIAT-PCIAT_04...6`4 `PCIAT-PCIAT_04...6`5 `PCIAT-PCIAT_04...6`Missing
## 0            0.00250941          0.0006273526                 0.001254705
## 1            0.04931507          0.0068493151                 0.001369863
## 2            0.16931217          0.0687830688                 0.005291005
## 3            0.20588235          0.5000000000                 0.000000000
##   `PCIAT-PCIAT_05...7`1 `PCIAT-PCIAT_05...7`2 `PCIAT-PCIAT_05...7`3
## 0           0.220828105           0.258469260             0.1066499
## 1           0.026027397           0.219178082             0.2917808
## 2           0.005291005           0.007936508             0.1005291
## 3           0.000000000           0.000000000             0.0000000
##   `PCIAT-PCIAT_05...7`4 `PCIAT-PCIAT_05...7`5 `PCIAT-PCIAT_05...7`Missing
## 0            0.03952321            0.01003764                 0.001882058
## 1            0.26438356            0.18904110                 0.004109589
## 2            0.33333333            0.53968254                 0.002645503
## 3            0.11764706            0.88235294                 0.000000000
```

```
##   `PCIAT-PCIAT_06...8`1 `PCIAT-PCIAT_06...8`2 `PCIAT-PCIAT_06...8`3
## 0            0.31493099            0.05332497           0.009410289
## 1            0.40136986            0.26301370           0.094520548
## 2            0.20370370            0.21693122           0.150793651
## 3            0.02941176            0.08823529           0.000000000
##   `PCIAT-PCIAT_06...8`4 `PCIAT-PCIAT_06...8`5 `PCIAT-PCIAT_06...8`Missing
## 0           0.003764115          0.0006273526                 0.001882058
## 1           0.045205479          0.0109589041                 0.000000000
## 2           0.224867725          0.1084656085                 0.002645503
## 3           0.294117647          0.5882352941                 0.000000000
##   `PCIAT-PCIAT_07...9`1 `PCIAT-PCIAT_07...9`2 `PCIAT-PCIAT_07...9`3
## 0            0.15307403            0.03324969           0.005018821
## 1            0.31506849            0.10821918           0.031506849
## 2            0.26190476            0.11375661           0.060846561
## 3            0.08823529            0.32352941           0.088235294
##   `PCIAT-PCIAT_07...9`4 `PCIAT-PCIAT_07...9`5 `PCIAT-PCIAT_07...9`Missing
## 0           0.001254705           0.000000000                 0.002509410
## 1           0.035616438           0.008219178                 0.001369863
## 2           0.111111111           0.055555556                 0.005291005
## 3           0.117647059           0.294117647                 0.000000000
##   `PCIAT-PCIAT_08...10`1 `PCIAT-PCIAT_08...10`2 `PCIAT-PCIAT_08...10`3
## 0              0.3023839             0.09598494             0.01442911
## 1              0.3328767             0.32602740             0.13835616
## 2              0.1084656             0.19841270             0.23280423
## 3              0.0000000             0.00000000             0.02941176
##   `PCIAT-PCIAT_08...10`4 `PCIAT-PCIAT_08...10`5 `PCIAT-PCIAT_08...10`Missing
## 0            0.004391468           0.0006273526                  0.002509410
## 1            0.065753425           0.0082191781                  0.002739726
## 2            0.309523810           0.0978835979                  0.000000000
## 3            0.205882353           0.7647058824                  0.000000000
##   `PCIAT-PCIAT_09...11`1 `PCIAT-PCIAT_09...11`2 `PCIAT-PCIAT_09...11`3
## 0             0.30175659             0.06148055            0.007528231
## 1             0.43013699             0.28904110            0.068493151
## 2             0.21957672             0.24338624            0.153439153
## 3             0.05882353             0.02941176            0.088235294
##   `PCIAT-PCIAT_09...11`4 `PCIAT-PCIAT_09...11`5 `PCIAT-PCIAT_09...11`Missing
## 0            0.001882058            0.001254705                  0.001882058
## 1            0.035616438            0.015068493                  0.004109589
## 2            0.193121693            0.126984127                  0.000000000
## 3            0.147058824            0.676470588                  0.000000000
##   `PCIAT-PCIAT_10...12`1 `PCIAT-PCIAT_10...12`2 `PCIAT-PCIAT_10...12`3
## 0             0.29736512              0.1317440              0.0119197
## 1             0.28493151              0.3917808              0.1424658
## 2             0.09523810              0.2301587              0.2010582
## 3             0.02941176              0.0000000              0.0000000
##   `PCIAT-PCIAT_10...12`4 `PCIAT-PCIAT_10...12`5 `PCIAT-PCIAT_10...12`Missing
## 0            0.006273526            0.000000000                  0.001254705
## 1            0.064383562            0.005479452                  0.001369863
## 2            0.330687831            0.095238095                  0.000000000
## 3            0.264705882            0.705882353                  0.000000000
##   `PCIAT-PCIAT_11...13`1 `PCIAT-PCIAT_11...13`2 `PCIAT-PCIAT_11...13`3
## 0             0.20953576              0.2208281             0.06398996
## 1             0.11780822              0.2876712             0.22739726
## 2             0.05820106              0.1031746             0.17724868
```

```
## 3             0.00000000              0.0000000             0.02941176
##   `PCIAT-PCIAT_11...13`4 `PCIAT-PCIAT_11...13`5 `PCIAT-PCIAT_11...13`Missing
## 0             0.01944793           0.0006273526                 0.0006273526
## 1             0.19589041           0.0465753425                 0.0000000000
## 2             0.42063492           0.1693121693                 0.0026455026
## 3             0.32352941           0.5000000000                 0.0000000000
##   `PCIAT-PCIAT_12...14`1 `PCIAT-PCIAT_12...14`2 `PCIAT-PCIAT_12...14`3
## 0              0.0978670           0.0006273526            0.000000000
## 1              0.3287671           0.0136986301            0.002739726
## 2              0.3544974           0.0449735450            0.007936508
## 3              0.2941176           0.1764705882            0.058823529
##   `PCIAT-PCIAT_12...14`4 `PCIAT-PCIAT_12...14`5 `PCIAT-PCIAT_12...14`Missing
## 0             0.00000000            0.000000000                 0.0006273526
## 1             0.00000000            0.000000000                 0.0027397260
## 2             0.01058201            0.005291005                 0.0052910053
## 3             0.05882353            0.029411765                 0.0000000000
##   `PCIAT-PCIAT_13...15`1 `PCIAT-PCIAT_13...15`2 `PCIAT-PCIAT_13...15`3
## 0              0.2735257             0.11668758             0.01693852
## 1              0.3260274             0.33150685             0.15342466
## 2              0.0978836             0.20370370             0.21428571
## 3              0.0000000             0.08823529             0.00000000
##   `PCIAT-PCIAT_13...15`4 `PCIAT-PCIAT_13...15`5 `PCIAT-PCIAT_13...15`Missing
## 0            0.006273526            0.001882058                  0.003136763
## 1            0.078082192            0.020547945                  0.001369863
## 2            0.293650794            0.169312169                  0.002645503
## 3            0.205882353            0.705882353                  0.000000000
##   `PCIAT-PCIAT_14...16`1 `PCIAT-PCIAT_14...16`2 `PCIAT-PCIAT_14...16`3
## 0              0.2547051             0.06085320            0.006273526
## 1              0.3547945             0.25890411            0.121917808
## 2              0.1084656             0.21428571            0.193121693
## 3              0.0000000             0.05882353            0.088235294
##   `PCIAT-PCIAT_14...16`4 `PCIAT-PCIAT_14...16`5 `PCIAT-PCIAT_14...16`Missing
## 0            0.001254705              0.0000000                  0.001254705
## 1            0.032876712              0.0109589                  0.001369863
## 2            0.246031746              0.1084656                  0.002645503
## 3            0.176470588              0.6764706                  0.000000000
##   `PCIAT-PCIAT_15...17`1 `PCIAT-PCIAT_15...17`2 `PCIAT-PCIAT_15...17`3
## 0             0.28042660             0.13048934             0.02258469
## 1             0.19315068             0.37808219             0.22328767
## 2             0.02910053             0.11375661             0.26455026
## 3             0.00000000             0.02941176             0.02941176
##   `PCIAT-PCIAT_15...17`4 `PCIAT-PCIAT_15...17`5 `PCIAT-PCIAT_15...17`Missing
## 0            0.004391468            0.001882058                  0.003136763
## 1            0.097260274            0.031506849                  0.000000000
## 2            0.365079365            0.206349206                  0.002645503
## 3            0.147058824            0.794117647                  0.000000000
##   `PCIAT-PCIAT_16...18`1 `PCIAT-PCIAT_16...18`2 `PCIAT-PCIAT_16...18`3
## 0             0.28795483             0.13801757             0.02321205
## 1             0.30547945             0.31780822             0.15205479
## 2             0.08201058             0.14285714             0.14285714
## 3             0.02941176             0.02941176             0.02941176
##   `PCIAT-PCIAT_16...18`4 `PCIAT-PCIAT_16...18`5 `PCIAT-PCIAT_16...18`Missing
## 0            0.008782936            0.001882058                  0.003136763
## 1            0.094520548            0.034246575                  0.002739726
```

```
## 2            0.325396825            0.269841270                   0.002645503
## 3            0.117647059            0.794117647                   0.000000000
##   `PCIAT-PCIAT_17...19`1 `PCIAT-PCIAT_17...19`2 `PCIAT-PCIAT_17...19`3
## 0             0.28481807              0.1875784             0.03638645
## 1             0.16712329              0.4109589             0.27260274
## 2             0.05026455              0.1058201             0.23809524
## 3             0.00000000              0.0000000             0.02941176
##   `PCIAT-PCIAT_17...19`4 `PCIAT-PCIAT_17...19`5 `PCIAT-PCIAT_17...19`Missing
## 0            0.008155583            0.001882058                   0.002509410
## 1            0.097260274            0.017808219                   0.002739726
## 2            0.396825397            0.185185185                   0.010582011
## 3            0.205882353            0.735294118                   0.029411765
##   `PCIAT-PCIAT_18`1 `PCIAT-PCIAT_18`2 `PCIAT-PCIAT_18`3 `PCIAT-PCIAT_18`4
## 0        0.30363864        0.16687578        0.02948557       0.007528231
## 1        0.21917808        0.38082192        0.18630137       0.110958904
## 2        0.04497354        0.10582011        0.17724868       0.328042328
## 3        0.00000000        0.02941176        0.00000000       0.058823529
##   `PCIAT-PCIAT_18`5 `PCIAT-PCIAT_18`Missing `PCIAT-PCIAT_19...21`1
## 0       0.003764115             0.002509410             0.29861982
## 1       0.039726027             0.004109589             0.36164384
## 2       0.325396825             0.002645503             0.14550265
## 3       0.911764706             0.000000000             0.05882353
##   `PCIAT-PCIAT_19...21`2 `PCIAT-PCIAT_19...21`3 `PCIAT-PCIAT_19...21`4
## 0             0.06775408            0.008782936             0.00250941
## 1             0.30136986            0.105479452             0.05479452
## 2             0.21693122            0.164021164             0.28042328
## 3             0.00000000            0.117647059             0.26470588
##   `PCIAT-PCIAT_19...21`5 `PCIAT-PCIAT_19...21`Missing `PCIAT-PCIAT_20...22`1
## 0            0.001882058                  0.001882058              0.2823087
## 1            0.021917808                  0.002739726              0.4780822
## 2            0.132275132                  0.002645503              0.2010582
## 3            0.558823529                  0.000000000              0.0000000
##   `PCIAT-PCIAT_20...22`2 `PCIAT-PCIAT_20...22`3 `PCIAT-PCIAT_20...22`4
## 0             0.03387704            0.001254705           0.0006273526
## 1             0.24931507            0.049315068           0.0273972603
## 2             0.27777778            0.164021164           0.2169312169
## 3             0.05882353            0.058823529           0.2941176471
##   `PCIAT-PCIAT_20...22`5 `PCIAT-PCIAT_20...22`Missing `PCIAT-PCIAT_Total`1
## 0           0.0006273526                 0.0006273526           0.02446675
## 1           0.0027397260                 0.0013698630           0.00000000
## 2           0.0767195767                 0.0026455026           0.00000000
## 3           0.5882352941                 0.0000000000           0.00000000
##   `PCIAT-PCIAT_Total`10 `PCIAT-PCIAT_Total`11 `PCIAT-PCIAT_Total`12
## 0             0.0250941            0.01819322            0.01819322
## 1             0.0000000            0.00000000            0.00000000
## 2             0.0000000            0.00000000            0.00000000
## 3             0.0000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`13 `PCIAT-PCIAT_Total`14 `PCIAT-PCIAT_Total`15
## 0            0.01819322            0.02885822            0.01568381
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`16 `PCIAT-PCIAT_Total`17 `PCIAT-PCIAT_Total`18
## 0            0.02321205            0.03074028            0.03074028
```

```
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`19 `PCIAT-PCIAT_Total`2 `PCIAT-PCIAT_Total`20
## 0            0.02823087           0.02634881            0.04265997
## 1            0.00000000           0.00000000            0.00000000
## 2            0.00000000           0.00000000            0.00000000
## 3            0.00000000           0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`21 `PCIAT-PCIAT_Total`22 `PCIAT-PCIAT_Total`23
## 0            0.03638645            0.03011292            0.03513174
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`24 `PCIAT-PCIAT_Total`25 `PCIAT-PCIAT_Total`26
## 0            0.03387704            0.03199498            0.03450439
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`27 `PCIAT-PCIAT_Total`28 `PCIAT-PCIAT_Total`29
## 0            0.03826851            0.03011292            0.03011292
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`3 `PCIAT-PCIAT_Total`30 `PCIAT-PCIAT_Total`31
## 0           0.01756587            0.03826851             0.0000000
## 1           0.00000000            0.00000000             0.0890411
## 2           0.00000000            0.00000000             0.0000000
## 3           0.00000000            0.00000000             0.0000000
##   `PCIAT-PCIAT_Total`32 `PCIAT-PCIAT_Total`33 `PCIAT-PCIAT_Total`34
## 0            0.00000000            0.00000000            0.00000000
## 1            0.06575342            0.05616438            0.07945205
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`35 `PCIAT-PCIAT_Total`36 `PCIAT-PCIAT_Total`37
## 0            0.00000000            0.00000000            0.00000000
## 1            0.05616438            0.06849315            0.04794521
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`38 `PCIAT-PCIAT_Total`39 `PCIAT-PCIAT_Total`4
## 0            0.00000000            0.00000000           0.01631117
## 1            0.03835616            0.05753425           0.00000000
## 2            0.00000000            0.00000000           0.00000000
## 3            0.00000000            0.00000000           0.00000000
##   `PCIAT-PCIAT_Total`40 `PCIAT-PCIAT_Total`41 `PCIAT-PCIAT_Total`42
## 0            0.00000000            0.00000000            0.00000000
## 1            0.05616438            0.04657534            0.06164384
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`43 `PCIAT-PCIAT_Total`44 `PCIAT-PCIAT_Total`45
## 0            0.00000000            0.00000000            0.00000000
## 1            0.04383562            0.04931507            0.04383562
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`46 `PCIAT-PCIAT_Total`47 `PCIAT-PCIAT_Total`48
```

```
## 0            0.00000000            0.00000000             0.0000000
## 1            0.03424658            0.04520548             0.0260274
## 2            0.00000000            0.00000000             0.0000000
## 3            0.00000000            0.00000000             0.0000000
##   `PCIAT-PCIAT_Total`49 `PCIAT-PCIAT_Total`5 `PCIAT-PCIAT_Total`50
## 0            0.00000000           0.01693852            0.00000000
## 1            0.03424658           0.00000000            0.00000000
## 2            0.00000000           0.00000000            0.05555556
## 3            0.00000000           0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`51 `PCIAT-PCIAT_Total`52 `PCIAT-PCIAT_Total`53
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.06878307            0.02380952            0.06349206
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`54 `PCIAT-PCIAT_Total`55 `PCIAT-PCIAT_Total`56
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.07142857            0.04497354            0.03968254
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`57 `PCIAT-PCIAT_Total`58 `PCIAT-PCIAT_Total`59
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.03703704            0.06084656            0.04761905
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`6 `PCIAT-PCIAT_Total`60 `PCIAT-PCIAT_Total`61
## 0           0.01756587            0.00000000            0.00000000
## 1           0.00000000            0.00000000            0.00000000
## 2           0.00000000            0.03703704            0.05026455
## 3           0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`62 `PCIAT-PCIAT_Total`63 `PCIAT-PCIAT_Total`64
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.03439153            0.01587302            0.04761905
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`65 `PCIAT-PCIAT_Total`66 `PCIAT-PCIAT_Total`67
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.03174603            0.03439153            0.02380952
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`68 `PCIAT-PCIAT_Total`69 `PCIAT-PCIAT_Total`7
## 0            0.00000000            0.00000000           0.02823087
## 1            0.00000000            0.00000000           0.00000000
## 2            0.02910053            0.01322751           0.00000000
## 3            0.00000000            0.00000000           0.00000000
##   `PCIAT-PCIAT_Total`70 `PCIAT-PCIAT_Total`71 `PCIAT-PCIAT_Total`72
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.02645503            0.01587302            0.02116402
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`73 `PCIAT-PCIAT_Total`74 `PCIAT-PCIAT_Total`75
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.01851852            0.02380952            0.01851852
## 3            0.00000000            0.00000000            0.00000000
```

```
##   `PCIAT-PCIAT_Total`76 `PCIAT-PCIAT_Total`77 `PCIAT-PCIAT_Total`78
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.01058201            0.01851852            0.01058201
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`79 `PCIAT-PCIAT_Total`8 `PCIAT-PCIAT_Total`80
## 0           0.000000000           0.01631117            0.00000000
## 1           0.000000000           0.00000000            0.00000000
## 2           0.005291005           0.00000000            0.00000000
## 3           0.000000000           0.00000000            0.08823529
##   `PCIAT-PCIAT_Total`81 `PCIAT-PCIAT_Total`82 `PCIAT-PCIAT_Total`83
## 0             0.0000000            0.00000000            0.00000000
## 1             0.0000000            0.00000000            0.00000000
## 2             0.0000000            0.00000000            0.00000000
## 3             0.1764706            0.02941176            0.08823529
##   `PCIAT-PCIAT_Total`84 `PCIAT-PCIAT_Total`85 `PCIAT-PCIAT_Total`87
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.02941176            0.05882353            0.08823529
##   `PCIAT-PCIAT_Total`88 `PCIAT-PCIAT_Total`89 `PCIAT-PCIAT_Total`9
## 0             0.0000000            0.00000000           0.02132999
## 1             0.0000000            0.00000000           0.00000000
## 2             0.0000000            0.00000000           0.00000000
## 3             0.1470588            0.08823529           0.00000000
##   `PCIAT-PCIAT_Total`90 `PCIAT-PCIAT_Total`91 `PCIAT-PCIAT_Total`92
## 0            0.00000000            0.00000000             0.0000000
## 1            0.00000000            0.00000000             0.0000000
## 2            0.00000000            0.00000000             0.0000000
## 3            0.02941176            0.02941176             0.1176471
##   `PCIAT-PCIAT_Total`93 `Basic_Demos-Age` `Basic_Demos-Sex` `CGAS-CGAS_Score`
## 0            0.00000000          9.254705         0.4002509          65.73965
## 1            0.00000000         11.117808         0.3410959          64.72603
## 2            0.00000000         12.291005         0.2698413          63.88889
## 3            0.02941176         14.676471         0.2352941          59.52941
##   `Physical-BMI` `Physical-Waist_Circumference` `Physical-Diastolic_BP`
## 0       18.28888                       25.91468                69.01129
## 1       19.64368                       26.37808                70.40000
## 2       20.31156                       26.33069                70.13492
## 3       26.01787                       27.08824                73.23529
##   `Physical-HeartRate` `Physical-Systolic_BP` `Fitness_Endurance-Max_Stage`
## 0             81.84253               115.1405                       5.030113
## 1             82.02055               118.6219                       4.961644
## 2             81.18783               119.3704                       4.957672
## 3             77.70588               129.5588                       4.911765
##   `Fitness_Endurance-Time_Mins` `Fitness_Endurance-Time_Sec` `FGC-FGC_CU`
## 0                      7.154956                     27.70514     9.634253
## 1                      7.058904                     28.61370    13.236986
## 2                      7.005291                     27.46561    14.354497
## 3                      6.852941                     27.23529    17.205882
##   `FGC-FGC_CU_Zone` `FGC-FGC_GSND_Zone` `FGC-FGC_GSD` `FGC-FGC_GSD_Zone`
## 0         0.6649937            1.955458      20.97176           1.966750
## 1         0.7041096            1.949315      21.70534           1.982192
## 2         0.6481481            1.878307      23.05757           1.910053
```

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## 3         0.6470588            1.970588      26.71176           2.088235
##   `FGC-FGC_PU` `FGC-FGC_PU_Zone` `FGC-FGC_SRL` `FGC-FGC_SRL_Zone`
## 0     4.252196         0.2277290      8.922867          0.7710163
## 1     5.669863         0.2575342      8.767945          0.7438356
## 2     6.171958         0.2460317      8.634788          0.6931217
## 3     7.941176         0.2058824      8.617647          0.5882353
##   `FGC-FGC_SRR_Zone` `FGC-FGC_TL` `FGC-FGC_TL_Zone`
## 0          0.7565872     8.871769         0.8419072
## 1          0.7575342     9.394110         0.8575342
## 2          0.7089947     9.347222         0.8306878
## 3          0.6470588     9.438235         0.7647059
##   `BIA-BIA_Activity_Level_num` `BIA-BIA_BMI` `BIA-BIA_BMR` `BIA-BIA_FFMI`
## 0                     2.748432      18.26980      1171.531       14.45546
## 1                     2.820548      19.07732      1191.249       14.59234
## 2                     2.822751      19.73794      1276.769       15.57741
## 3                     3.088235      24.55606      1460.271       16.99771
##   `BIA-BIA_FMI` `BIA-BIA_Frame_num` `PAQ_A-PAQ_A_Total` `PAQ_C-PAQ_C_Total`
## 0      3.765062            1.764115            2.094741            2.567085
## 1      4.434212            1.883562            2.088767            2.577727
## 2      4.108381            1.923280            2.098952            2.578786
## 3      7.514756            2.205882            2.119088            2.553529
##   `PCIAT-PCIAT_01...55` `PCIAT-PCIAT_02...56` `PCIAT-PCIAT_03...57`
## 0              1.528858              1.210163              1.490590
## 1              3.168493              3.127397              3.302740
## 2              4.166667              4.174603              4.256614
## 3              4.705882              4.941176              4.941176
##   `PCIAT-PCIAT_04...58` `PCIAT-PCIAT_05...59` `PCIAT-PCIAT_06...60`
## 0             0.3375157              1.269762             0.4698871
## 1             1.1383562              3.350685             1.4465753
## 2             2.0873016              4.359788             2.5343915
## 3             3.9411765              4.882353             4.3235294
##   `PCIAT-PCIAT_07...61` `PCIAT-PCIAT_08...62` `PCIAT-PCIAT_09...63`
## 0             0.2396487             0.5608532             0.4629862
## 1             0.8095890             1.7068493             1.4356164
## 2             1.3941799             2.9312169             2.5740741
## 3             2.9411765             4.7352941             4.3529412
##   `PCIAT-PCIAT_10...64` `PCIAT-PCIAT_11...65` `PCIAT-PCIAT_12...66`
## 0             0.6229611              0.925345            0.09912171
## 1             1.7821918              2.391781            0.36438356
## 2             2.9576720              3.330688            0.53703704
## 3             4.6176471              3.882353            1.20588235
##   `PCIAT-PCIAT_13...67` `PCIAT-PCIAT_14...68` `PCIAT-PCIAT_15...69`
## 0             0.5953576             0.4015056             0.6392723
## 1             1.8657534             1.4260274             2.1657534
## 2             3.1719577             2.6455026             3.5449735
## 3             4.5294118             4.4705882             4.7058824
##   `PCIAT-PCIAT_16...70` `PCIAT-PCIAT_17...71` `PCIAT-PCIAT_19...72`
## 0             0.6813049             0.8136763             0.4818068
## 1             1.9493151             2.2876712             1.6123288
## 2             3.4497354             3.5000000             2.8571429
## 3             4.6176471             4.6176471             4.2647059
##   `PCIAT-PCIAT_20...73` `SDS-SDS_Total_T`
## 0             0.3601004          55.37453
## 1             1.2493151          58.93014
```

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## 2             2.5026455          62.34392
## 3             4.4117647          68.02941
##   `PreInt_EduHx-computerinternet_hoursday`
## 0                                0.7390213
## 1                                1.2493151
## 2                                1.6269841
## 3                                2.0882353
##
## Coefficients of linear discriminants:
##                                                    LD1           LD2
## `PCIAT-PCIAT_01...3`1                    -6.650459e-01  2.209243e-01
## `PCIAT-PCIAT_01...3`2                    -4.926259e-01  3.176756e-01
## `PCIAT-PCIAT_01...3`3                    -2.454976e-01  7.645130e-02
## `PCIAT-PCIAT_01...3`4                     1.774063e-01 -1.105148e-01
## `PCIAT-PCIAT_01...3`5                     5.825163e-01 -3.895716e-02
## `PCIAT-PCIAT_01...3`Missing              -2.775832e+00 -5.532813e+00
## `PCIAT-PCIAT_02...4`1                    -1.350673e-01  2.118350e-02
## `PCIAT-PCIAT_02...4`2                    -1.849285e-01  4.364813e-02
## `PCIAT-PCIAT_02...4`3                     2.561878e-02 -1.758761e-01
## `PCIAT-PCIAT_02...4`4                     2.229393e-01 -3.336279e-01
## `PCIAT-PCIAT_02...4`5                     6.835400e-01  8.617630e-02
## `PCIAT-PCIAT_02...4`Missing               8.609587e+00  1.112641e+01
## `PCIAT-PCIAT_03...5`1                    -1.397051e+00  5.470837e-01
## `PCIAT-PCIAT_03...5`2                    -1.231930e+00  4.759142e-01
## `PCIAT-PCIAT_03...5`3                    -6.679335e-01  8.042655e-02
## `PCIAT-PCIAT_03...5`4                     7.840066e-02 -1.629200e-01
## `PCIAT-PCIAT_03...5`5                     7.840904e-01 -5.307874e-01
## `PCIAT-PCIAT_03...5`Missing              -1.484183e+00 -1.405166e+00
## `PCIAT-PCIAT_04...6`1                     2.042245e-02 -1.909035e-01
## `PCIAT-PCIAT_04...6`2                     1.934882e-01 -2.442210e-01
## `PCIAT-PCIAT_04...6`3                     4.147650e-01  2.460701e-01
## `PCIAT-PCIAT_04...6`4                     6.394558e-01 -1.014362e-01
## `PCIAT-PCIAT_04...6`5                     1.353592e+00  2.036682e+00
## `PCIAT-PCIAT_04...6`Missing               2.888647e-01  6.136239e-01
## `PCIAT-PCIAT_05...7`1                    -6.820819e-01  5.416285e-01
## `PCIAT-PCIAT_05...7`2                    -5.865382e-01  2.940980e-01
## `PCIAT-PCIAT_05...7`3                    -2.296015e-01 -7.713189e-02
## `PCIAT-PCIAT_05...7`4                     1.766342e-01 -2.688400e-01
## `PCIAT-PCIAT_05...7`5                     8.270200e-01 -3.689418e-01
## `PCIAT-PCIAT_05...7`Missing              -1.338101e+00 -1.738281e+00
## `PCIAT-PCIAT_06...8`1                     1.136894e-02 -1.578377e-01
## `PCIAT-PCIAT_06...8`2                     2.466760e-01 -2.796639e-01
## `PCIAT-PCIAT_06...8`3                     4.253437e-01 -3.956968e-01
## `PCIAT-PCIAT_06...8`4                     6.521134e-01  6.204581e-01
## `PCIAT-PCIAT_06...8`5                     8.282084e-01  1.319725e+00
## `PCIAT-PCIAT_06...8`Missing               1.233111e-01  9.109240e-01
## `PCIAT-PCIAT_07...9`1                     1.260426e-01 -9.199690e-02
## `PCIAT-PCIAT_07...9`2                     4.654143e-01  6.924003e-02
## `PCIAT-PCIAT_07...9`3                     3.489940e-01  7.388064e-02
## `PCIAT-PCIAT_07...9`4                     7.586854e-01  8.157897e-02
## `PCIAT-PCIAT_07...9`5                     8.937860e-01  6.348154e-01
## `PCIAT-PCIAT_07...9`Missing              -4.451629e-01 -1.559620e+00
## `PCIAT-PCIAT_08...10`1                   -1.381490e-01 -2.029532e-01
## `PCIAT-PCIAT_08...10`2                    1.036696e-01 -3.685036e-01
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## `PCIAT-PCIAT_08...10`3                    2.957812e-01 -4.066840e-01
## `PCIAT-PCIAT_08...10`4                    4.535599e-01 -8.236472e-02
## `PCIAT-PCIAT_08...10`5                    9.586880e-01  2.582145e+00
## `PCIAT-PCIAT_08...10`Missing             -1.158881e+00 -3.728032e-01
## `PCIAT-PCIAT_09...11`1                    5.335644e-02 -2.477570e-01
## `PCIAT-PCIAT_09...11`2                    2.321013e-01 -3.956336e-01
## `PCIAT-PCIAT_09...11`3                    4.314345e-01 -9.499101e-02
## `PCIAT-PCIAT_09...11`4                    5.021345e-01 -6.046888e-02
## `PCIAT-PCIAT_09...11`5                    7.606529e-01  1.020559e+00
## `PCIAT-PCIAT_09...11`Missing             -1.280110e-01 -1.362175e+00
## `PCIAT-PCIAT_10...12`1                   -2.273468e-01 -1.140571e-01
## `PCIAT-PCIAT_10...12`2                    4.728492e-02 -2.457658e-01
## `PCIAT-PCIAT_10...12`3                    3.414327e-01 -4.505429e-01
## `PCIAT-PCIAT_10...12`4                    5.463968e-01  1.617771e-01
## `PCIAT-PCIAT_10...12`5                    1.074800e+00  1.973160e+00
## `PCIAT-PCIAT_10...12`Missing             -1.342293e+00  1.918912e-01
## `PCIAT-PCIAT_11...13`1                   -2.922677e-01  2.329790e-01
## `PCIAT-PCIAT_11...13`2                   -1.398789e-01  2.748357e-02
## `PCIAT-PCIAT_11...13`3                    1.044965e-01 -1.082171e-01
## `PCIAT-PCIAT_11...13`4                    4.248826e-01 -1.230810e-01
## `PCIAT-PCIAT_11...13`5                    5.604158e-01 -3.198139e-01
## `PCIAT-PCIAT_11...13`Missing             -1.198686e+00 -4.242255e-01
## `PCIAT-PCIAT_12...14`1                    3.824234e-01 -2.240519e-01
## `PCIAT-PCIAT_12...14`2                    2.612488e-01  2.600279e-01
## `PCIAT-PCIAT_12...14`3                    5.158172e-01  6.719813e-01
## `PCIAT-PCIAT_12...14`4                    1.040586e+00  4.004728e+00
## `PCIAT-PCIAT_12...14`5                    1.019758e+00  1.447154e+00
## `PCIAT-PCIAT_12...14`Missing             -1.359947e-01 -1.517630e+00
## `PCIAT-PCIAT_13...15`1                   -1.710140e-01 -2.028259e-01
## `PCIAT-PCIAT_13...15`2                   -4.096238e-02 -1.824231e-01
## `PCIAT-PCIAT_13...15`3                    2.354302e-01 -3.580175e-01
## `PCIAT-PCIAT_13...15`4                    6.167926e-01  7.127890e-02
## `PCIAT-PCIAT_13...15`5                    7.469073e-01  4.955972e-01
## `PCIAT-PCIAT_13...15`Missing             -4.269435e-02 -4.722917e-02
## `PCIAT-PCIAT_14...16`1                    9.901037e-02 -2.108175e-01
## `PCIAT-PCIAT_14...16`2                    2.183239e-01 -3.877561e-01
## `PCIAT-PCIAT_14...16`3                    3.577382e-01 -4.178335e-01
## `PCIAT-PCIAT_14...16`4                    6.705040e-01  2.860089e-01
## `PCIAT-PCIAT_14...16`5                    1.022664e+00  1.945464e+00
## `PCIAT-PCIAT_14...16`Missing              5.669740e-02  5.082576e-01
## `PCIAT-PCIAT_15...17`1                   -3.775589e-01  1.017636e-01
## `PCIAT-PCIAT_15...17`2                   -1.537020e-01 -1.027503e-01
## `PCIAT-PCIAT_15...17`3                    2.018023e-01 -3.022504e-01
## `PCIAT-PCIAT_15...17`4                    5.221311e-01 -2.120428e-01
## `PCIAT-PCIAT_15...17`5                    6.691834e-01  4.003094e-01
## `PCIAT-PCIAT_15...17`Missing             -7.393327e-01  1.997470e-02
## `PCIAT-PCIAT_16...18`1                   -9.973951e-02  8.795519e-02
## `PCIAT-PCIAT_16...18`2                   -7.108652e-02 -3.550526e-02
## `PCIAT-PCIAT_16...18`3                    1.505924e-01 -3.524297e-03
## `PCIAT-PCIAT_16...18`4                    3.345675e-01 -2.110760e-01
## `PCIAT-PCIAT_16...18`5                    6.884023e-01  1.283432e-02
## `PCIAT-PCIAT_16...18`Missing             -6.561404e-01 -6.169654e-01
## `PCIAT-PCIAT_17...19`1                   -4.587573e-01 -1.177985e-02
## `PCIAT-PCIAT_17...19`2                   -2.726581e-01 -1.920421e-01
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## `PCIAT-PCIAT_17...19`3                    3.933181e-02 -2.995511e-01
## `PCIAT-PCIAT_17...19`4                    4.531532e-01 -1.917904e-02
## `PCIAT-PCIAT_17...19`5                    8.144806e-01  5.328986e-01
## `PCIAT-PCIAT_17...19`Missing             -2.256016e-01  1.219359e+00
## `PCIAT-PCIAT_18`1                        -5.557615e-01  3.931266e-01
## `PCIAT-PCIAT_18`2                        -2.658838e-01  1.965192e-01
## `PCIAT-PCIAT_18`3                         1.541814e-02  1.827089e-01
## `PCIAT-PCIAT_18`4                         4.505185e-01  1.186852e-01
## `PCIAT-PCIAT_18`5                         9.773009e-01  9.976137e-01
## `PCIAT-PCIAT_18`Missing                  -5.013644e-01 -1.258384e+00
## `PCIAT-PCIAT_19...21`1                    2.097716e-02 -8.322713e-02
## `PCIAT-PCIAT_19...21`2                    1.750240e-01 -2.760732e-01
## `PCIAT-PCIAT_19...21`3                    4.469753e-01  1.062920e-01
## `PCIAT-PCIAT_19...21`4                    6.179897e-01  6.600183e-02
## `PCIAT-PCIAT_19...21`5                    7.193755e-01  2.097588e-01
## `PCIAT-PCIAT_19...21`Missing              8.634123e-02  5.016244e-01
## `PCIAT-PCIAT_20...22`1                    9.619848e-02 -3.698413e-01
## `PCIAT-PCIAT_20...22`2                    2.921709e-01 -5.781617e-01
## `PCIAT-PCIAT_20...22`3                    4.565035e-01 -4.388744e-01
## `PCIAT-PCIAT_20...22`4                    6.188865e-01  3.506354e-01
## `PCIAT-PCIAT_20...22`5                    9.000991e-01  2.669803e+00
## `PCIAT-PCIAT_20...22`Missing              1.026462e+00  3.263484e-01
## `PCIAT-PCIAT_Total`1                      1.220278e+00 -4.195999e-01
## `PCIAT-PCIAT_Total`10                     3.043001e+00 -8.487212e-01
## `PCIAT-PCIAT_Total`11                     2.364392e+00 -5.250705e-01
## `PCIAT-PCIAT_Total`12                     2.912018e+00 -7.864389e-01
## `PCIAT-PCIAT_Total`13                     2.606723e+00 -5.452097e-01
## `PCIAT-PCIAT_Total`14                     2.638093e+00 -4.483066e-01
## `PCIAT-PCIAT_Total`15                     2.367137e+00 -5.985469e-01
## `PCIAT-PCIAT_Total`16                     2.634309e+00 -4.021325e-01
## `PCIAT-PCIAT_Total`17                     2.267661e+00 -3.511316e-01
## `PCIAT-PCIAT_Total`18                     2.269956e+00 -1.967270e-01
## `PCIAT-PCIAT_Total`19                     2.163851e+00 -1.896712e-01
## `PCIAT-PCIAT_Total`2                      1.489707e+00 -4.988308e-01
## `PCIAT-PCIAT_Total`20                     1.879751e+00 -7.693446e-02
## `PCIAT-PCIAT_Total`21                     1.744064e+00  1.603392e-02
## `PCIAT-PCIAT_Total`22                     1.573803e+00 -7.546240e-02
## `PCIAT-PCIAT_Total`23                     1.596727e+00  1.645526e-01
## `PCIAT-PCIAT_Total`24                     9.890027e-01  4.144476e-01
## `PCIAT-PCIAT_Total`25                     8.095825e-01  3.019367e-01
## `PCIAT-PCIAT_Total`26                     6.818295e-01  5.697143e-01
## `PCIAT-PCIAT_Total`27                     3.190740e-01  9.408366e-01
## `PCIAT-PCIAT_Total`28                    -7.993663e-02  1.074006e+00
## `PCIAT-PCIAT_Total`29                    -5.603134e-01  8.991416e-01
## `PCIAT-PCIAT_Total`3                      2.022260e+00 -6.692900e-01
## `PCIAT-PCIAT_Total`30                    -6.075393e-01  1.119519e+00
## `PCIAT-PCIAT_Total`31                     3.021273e+00 -7.949776e-01
## `PCIAT-PCIAT_Total`32                     2.545995e+00 -6.499374e-01
## `PCIAT-PCIAT_Total`33                     2.310403e+00 -3.255158e-01
## `PCIAT-PCIAT_Total`34                     1.923638e+00 -4.517283e-01
## `PCIAT-PCIAT_Total`35                     1.319243e+00 -2.908909e-01
## `PCIAT-PCIAT_Total`36                     9.851728e-01 -2.802049e-01
## `PCIAT-PCIAT_Total`37                     9.700287e-01  1.509431e-01
## `PCIAT-PCIAT_Total`38                     7.264513e-01  1.568194e-01
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## `PCIAT-PCIAT_Total`39                    -9.823338e-02  4.965542e-01
## `PCIAT-PCIAT_Total`4                      1.926829e+00 -7.249785e-01
## `PCIAT-PCIAT_Total`40                    -4.353287e-01  2.338984e-01
## `PCIAT-PCIAT_Total`41                    -8.472079e-01  5.683713e-01
## `PCIAT-PCIAT_Total`42                    -1.228676e+00  3.510779e-01
## `PCIAT-PCIAT_Total`43                    -1.772715e+00 -3.069378e-02
## `PCIAT-PCIAT_Total`44                    -2.209839e+00 -3.598442e-02
## `PCIAT-PCIAT_Total`45                    -2.509660e+00  5.378179e-01
## `PCIAT-PCIAT_Total`46                    -2.715749e+00  3.452114e-01
## `PCIAT-PCIAT_Total`47                    -3.141312e+00  8.065392e-01
## `PCIAT-PCIAT_Total`48                    -3.556494e+00  4.833276e-01
## `PCIAT-PCIAT_Total`49                    -4.013476e+00  3.750844e-01
## `PCIAT-PCIAT_Total`5                      3.000381e+00 -9.948962e-01
## `PCIAT-PCIAT_Total`50                     4.814537e+00  2.353347e+00
## `PCIAT-PCIAT_Total`51                     4.160240e+00  1.497246e+00
## `PCIAT-PCIAT_Total`52                     4.031799e+00  1.802649e+00
## `PCIAT-PCIAT_Total`53                     3.614458e+00  1.998707e+00
## `PCIAT-PCIAT_Total`54                     2.808615e+00  1.053060e+00
## `PCIAT-PCIAT_Total`55                     2.515560e+00  1.556603e+00
## `PCIAT-PCIAT_Total`56                     2.183268e+00  7.105684e-01
## `PCIAT-PCIAT_Total`57                     1.655573e+00  3.857256e-01
## `PCIAT-PCIAT_Total`58                     1.170363e+00  6.399079e-01
## `PCIAT-PCIAT_Total`59                     1.185241e+00  1.109319e+00
## `PCIAT-PCIAT_Total`6                      2.653969e+00 -9.522457e-01
## `PCIAT-PCIAT_Total`60                     2.637636e-01  1.128541e-01
## `PCIAT-PCIAT_Total`61                    -4.727155e-02  7.243999e-01
## `PCIAT-PCIAT_Total`62                    -3.821975e-01  1.144603e+00
## `PCIAT-PCIAT_Total`63                    -7.808281e-01  3.126210e-01
## `PCIAT-PCIAT_Total`64                    -1.285528e+00 -3.397566e-01
## `PCIAT-PCIAT_Total`65                    -1.658210e+00  6.892320e-01
## `PCIAT-PCIAT_Total`66                    -2.181231e+00 -1.022094e+00
## `PCIAT-PCIAT_Total`67                    -2.847216e+00 -2.207807e+00
## `PCIAT-PCIAT_Total`68                    -3.156031e+00 -2.439664e-01
## `PCIAT-PCIAT_Total`69                    -3.772846e+00 -9.543065e-01
## `PCIAT-PCIAT_Total`7                      2.859900e+00 -9.502997e-01
## `PCIAT-PCIAT_Total`70                    -3.978942e+00 -1.564237e+00
## `PCIAT-PCIAT_Total`71                    -4.859762e+00 -2.954636e+00
## `PCIAT-PCIAT_Total`72                    -4.982994e+00 -1.869339e+00
## `PCIAT-PCIAT_Total`73                    -5.326591e+00 -3.011906e+00
## `PCIAT-PCIAT_Total`74                    -6.139159e+00 -5.719430e+00
## `PCIAT-PCIAT_Total`75                    -6.726316e+00 -5.994889e+00
## `PCIAT-PCIAT_Total`76                    -7.263747e+00 -5.362048e+00
## `PCIAT-PCIAT_Total`77                    -7.147640e+00 -3.276476e+00
## `PCIAT-PCIAT_Total`78                    -8.182445e+00 -4.780466e+00
## `PCIAT-PCIAT_Total`79                    -8.715825e+00 -7.535422e+00
## `PCIAT-PCIAT_Total`8                      3.001948e+00 -1.105979e+00
## `PCIAT-PCIAT_Total`80                     2.951543e+00  4.313870e+00
## `PCIAT-PCIAT_Total`81                     2.499241e+00  3.855002e+00
## `PCIAT-PCIAT_Total`82                     4.623220e-01 -3.920563e+00
## `PCIAT-PCIAT_Total`83                     1.578491e+00  2.538845e+00
## `PCIAT-PCIAT_Total`84                     7.851586e-01 -2.909874e-02
## `PCIAT-PCIAT_Total`85                     6.056495e-01  1.417927e+00
## `PCIAT-PCIAT_Total`87                    -1.155307e+00 -3.286291e+00
## `PCIAT-PCIAT_Total`88                    -7.423395e-01  3.348299e-01
```

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## `PCIAT-PCIAT_Total`89                    -1.575949e+00 -3.097628e+00
## `PCIAT-PCIAT_Total`9                      3.489534e+00 -1.117959e+00
## `PCIAT-PCIAT_Total`90                    -2.306134e+00 -4.799391e+00
## `PCIAT-PCIAT_Total`91                    -1.898427e+00 -2.425676e+00
## `PCIAT-PCIAT_Total`92                    -2.368320e+00 -2.172485e+00
## `PCIAT-PCIAT_Total`93                    -4.136337e+00 -6.093089e+00
## `Basic_Demos-Age`                        -5.332064e-02  3.232656e-03
## `Basic_Demos-Sex`                         2.042874e-02  1.820890e-02
## `CGAS-CGAS_Score`                         7.660285e-04  2.353469e-03
## `Physical-BMI`                           -1.849765e-03 -1.458115e-02
## `Physical-Waist_Circumference`           -3.968752e-03 -2.034730e-02
## `Physical-Diastolic_BP`                  -6.148815e-04 -6.516979e-04
## `Physical-HeartRate`                     -4.151844e-04 -1.720397e-03
## `Physical-Systolic_BP`                    4.827451e-06  1.033062e-03
## `Fitness_Endurance-Max_Stage`            -4.594201e-03  5.943152e-02
## `Fitness_Endurance-Time_Mins`            -5.959666e-03 -2.928685e-02
## `Fitness_Endurance-Time_Sec`              2.138395e-04 -3.221452e-03
## `FGC-FGC_CU`                              3.919053e-03 -6.969427e-03
## `FGC-FGC_CU_Zone`                        -3.784820e-02  5.882880e-02
## `FGC-FGC_GSND_Zone`                      -7.504367e-02 -2.034934e-02
## `FGC-FGC_GSD`                            -2.963022e-03 -1.233886e-02
## `FGC-FGC_GSD_Zone`                        7.124476e-02  7.036395e-02
## `FGC-FGC_PU`                             -3.081277e-04  9.579014e-03
## `FGC-FGC_PU_Zone`                         9.207118e-02 -3.416022e-02
## `FGC-FGC_SRL`                             8.148005e-03  5.254671e-03
## `FGC-FGC_SRL_Zone`                       -1.135967e-01  2.544325e-02
## `FGC-FGC_SRR_Zone`                        1.234549e-01 -1.373312e-01
## `FGC-FGC_TL`                             -6.520280e-03 -1.989197e-02
## `FGC-FGC_TL_Zone`                         3.029634e-02  1.629388e-01
## `BIA-BIA_Activity_Level_num`              1.858356e-02  6.006304e-02
## `BIA-BIA_BMI`                            -2.095776e-02  3.698698e-01
## `BIA-BIA_BMR`                             2.579636e-05  4.173805e-06
## `BIA-BIA_FFMI`                            3.021423e-02 -3.489489e-01
## `BIA-BIA_FMI`                             3.283049e-02 -3.311590e-01
## `BIA-BIA_Frame_num`                      -4.994184e-02 -8.941850e-02
## `PAQ_A-PAQ_A_Total`                       8.552622e-02  9.280285e-02
## `PAQ_C-PAQ_C_Total`                       6.794167e-02 -2.541965e-02
## `PCIAT-PCIAT_01...55`                     5.617593e-03  4.088324e-02
## `PCIAT-PCIAT_02...56`                     2.003605e-01 -1.024309e-01
## `PCIAT-PCIAT_03...57`                    -4.254684e-01 -1.089562e-02
## `PCIAT-PCIAT_04...58`                     2.915259e-01  8.296791e-02
## `PCIAT-PCIAT_05...59`                     4.878161e-02 -1.150973e-01
## `PCIAT-PCIAT_06...60`                     3.224086e-01  9.013494e-02
## `PCIAT-PCIAT_07...61`                     2.651137e-01  5.172139e-02
## `PCIAT-PCIAT_08...62`                     2.471477e-01 -1.523019e-02
## `PCIAT-PCIAT_09...63`                     3.051590e-01 -5.614442e-02
## `PCIAT-PCIAT_10...64`                     2.531954e-01  1.157284e-02
## `PCIAT-PCIAT_11...65`                     1.318109e-01 -6.342489e-02
## `PCIAT-PCIAT_12...66`                     3.411051e-01  8.898859e-02
## `PCIAT-PCIAT_13...67`                     2.427135e-01 -7.687169e-02
## `PCIAT-PCIAT_14...68`                     3.507389e-01  3.894911e-02
## `PCIAT-PCIAT_15...69`                     1.888917e-01 -7.617964e-02
## `PCIAT-PCIAT_16...70`                     1.774983e-01 -3.857141e-02
## `PCIAT-PCIAT_17...71`                     7.954686e-02 -7.844181e-02
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## `PCIAT-PCIAT_19...72`                     3.311440e-01 -2.586526e-02
## `PCIAT-PCIAT_20...73`                     3.864486e-01  5.738453e-04
## `SDS-SDS_Total_T`                        -4.953538e-03 -2.782425e-03
## `PreInt_EduHx-computerinternet_hoursday` -1.198041e-02  2.015312e-02
##                                                     LD3
## `PCIAT-PCIAT_01...3`1                    -1.088116e-01
## `PCIAT-PCIAT_01...3`2                    -3.496644e-01
## `PCIAT-PCIAT_01...3`3                    -9.095710e-02
## `PCIAT-PCIAT_01...3`4                     1.310095e-01
## `PCIAT-PCIAT_01...3`5                     3.274965e-02
## `PCIAT-PCIAT_01...3`Missing               4.230122e+00
## `PCIAT-PCIAT_02...4`1                    -8.278624e-02
## `PCIAT-PCIAT_02...4`2                     1.516145e-02
## `PCIAT-PCIAT_02...4`3                     4.263165e-01
## `PCIAT-PCIAT_02...4`4                     4.594698e-02
## `PCIAT-PCIAT_02...4`5                    -1.727025e-01
## `PCIAT-PCIAT_02...4`Missing              -3.905348e+00
## `PCIAT-PCIAT_03...5`1                    -4.668222e-01
## `PCIAT-PCIAT_03...5`2                    -2.647897e-01
## `PCIAT-PCIAT_03...5`3                     2.892116e-01
## `PCIAT-PCIAT_03...5`4                     2.076560e-01
## `PCIAT-PCIAT_03...5`5                    -3.819537e-01
## `PCIAT-PCIAT_03...5`Missing               1.193846e+00
## `PCIAT-PCIAT_04...6`1                     1.940365e-02
## `PCIAT-PCIAT_04...6`2                    -1.054151e-01
## `PCIAT-PCIAT_04...6`3                    -2.318983e-01
## `PCIAT-PCIAT_04...6`4                    -7.114381e-01
## `PCIAT-PCIAT_04...6`5                     1.574288e+00
## `PCIAT-PCIAT_04...6`Missing              -1.841755e+00
## `PCIAT-PCIAT_05...7`1                    -3.424971e-01
## `PCIAT-PCIAT_05...7`2                     3.353236e-02
## `PCIAT-PCIAT_05...7`3                     3.495105e-01
## `PCIAT-PCIAT_05...7`4                     3.421271e-01
## `PCIAT-PCIAT_05...7`5                    -1.940244e-01
## `PCIAT-PCIAT_05...7`Missing               1.621368e+00
## `PCIAT-PCIAT_06...8`1                     4.558315e-02
## `PCIAT-PCIAT_06...8`2                     3.203758e-02
## `PCIAT-PCIAT_06...8`3                    -1.207071e-01
## `PCIAT-PCIAT_06...8`4                    -4.617482e-01
## `PCIAT-PCIAT_06...8`5                     3.184192e-01
## `PCIAT-PCIAT_06...8`Missing              -2.419102e-02
## `PCIAT-PCIAT_07...9`1                     9.364210e-02
## `PCIAT-PCIAT_07...9`2                     3.342095e-01
## `PCIAT-PCIAT_07...9`3                    -1.134378e-01
## `PCIAT-PCIAT_07...9`4                    -2.553341e-01
## `PCIAT-PCIAT_07...9`5                    -1.330732e-01
## `PCIAT-PCIAT_07...9`Missing              -6.197611e-01
## `PCIAT-PCIAT_08...10`1                    1.462967e-01
## `PCIAT-PCIAT_08...10`2                    1.261121e-02
## `PCIAT-PCIAT_08...10`3                   -4.644729e-01
## `PCIAT-PCIAT_08...10`4                   -5.445270e-01
## `PCIAT-PCIAT_08...10`5                    2.047907e+00
## `PCIAT-PCIAT_08...10`Missing              1.640928e-02
## `PCIAT-PCIAT_09...11`1                    1.599796e-01
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## `PCIAT-PCIAT_09...11`2                    2.670488e-01
## `PCIAT-PCIAT_09...11`3                   -2.898651e-01
## `PCIAT-PCIAT_09...11`4                   -6.798030e-01
## `PCIAT-PCIAT_09...11`5                    2.876071e-01
## `PCIAT-PCIAT_09...11`Missing              2.555918e+00
## `PCIAT-PCIAT_10...12`1                    1.810179e-01
## `PCIAT-PCIAT_10...12`2                    2.361105e-01
## `PCIAT-PCIAT_10...12`3                   -1.668083e-01
## `PCIAT-PCIAT_10...12`4                   -8.186000e-01
## `PCIAT-PCIAT_10...12`5                    1.265491e+00
## `PCIAT-PCIAT_10...12`Missing             -1.449828e+00
## `PCIAT-PCIAT_11...13`1                   -2.011607e-01
## `PCIAT-PCIAT_11...13`2                    4.523658e-03
## `PCIAT-PCIAT_11...13`3                    1.057574e-01
## `PCIAT-PCIAT_11...13`4                    3.906214e-02
## `PCIAT-PCIAT_11...13`5                   -1.226468e-01
## `PCIAT-PCIAT_11...13`Missing             -1.321494e+00
## `PCIAT-PCIAT_12...14`1                   -2.876865e-05
## `PCIAT-PCIAT_12...14`2                    3.134664e-01
## `PCIAT-PCIAT_12...14`3                    8.273338e-01
## `PCIAT-PCIAT_12...14`4                    4.281376e-01
## `PCIAT-PCIAT_12...14`5                    3.215840e-02
## `PCIAT-PCIAT_12...14`Missing             -2.241036e+00
## `PCIAT-PCIAT_13...15`1                    1.605466e-01
## `PCIAT-PCIAT_13...15`2                    2.360268e-01
## `PCIAT-PCIAT_13...15`3                    2.853447e-02
## `PCIAT-PCIAT_13...15`4                   -3.141141e-01
## `PCIAT-PCIAT_13...15`5                   -5.676684e-02
## `PCIAT-PCIAT_13...15`Missing              7.911125e-03
## `PCIAT-PCIAT_14...16`1                    1.939943e-01
## `PCIAT-PCIAT_14...16`2                    3.783765e-02
## `PCIAT-PCIAT_14...16`3                   -2.976215e-01
## `PCIAT-PCIAT_14...16`4                   -8.488988e-01
## `PCIAT-PCIAT_14...16`5                    1.377015e+00
## `PCIAT-PCIAT_14...16`Missing              4.956923e-02
## `PCIAT-PCIAT_15...17`1                    3.287462e-02
## `PCIAT-PCIAT_15...17`2                    3.931232e-01
## `PCIAT-PCIAT_15...17`3                    1.138181e-01
## `PCIAT-PCIAT_15...17`4                   -5.421667e-01
## `PCIAT-PCIAT_15...17`5                   -2.546621e-02
## `PCIAT-PCIAT_15...17`Missing             -1.903757e+00
## `PCIAT-PCIAT_16...18`1                    1.212298e-01
## `PCIAT-PCIAT_16...18`2                    1.587428e-01
## `PCIAT-PCIAT_16...18`3                    3.406770e-01
## `PCIAT-PCIAT_16...18`4                   -2.205373e-01
## `PCIAT-PCIAT_16...18`5                   -2.346127e-01
## `PCIAT-PCIAT_16...18`Missing              6.629042e-01
## `PCIAT-PCIAT_17...19`1                    6.428754e-02
## `PCIAT-PCIAT_17...19`2                    3.991590e-01
## `PCIAT-PCIAT_17...19`3                    3.555402e-01
## `PCIAT-PCIAT_17...19`4                   -4.041597e-01
## `PCIAT-PCIAT_17...19`5                   -4.059367e-01
## `PCIAT-PCIAT_17...19`Missing              7.493362e-01
## `PCIAT-PCIAT_18`1                        -4.182326e-01
```

```
## `PCIAT-PCIAT_18`2                        -2.491931e-01
## `PCIAT-PCIAT_18`3                        -4.522254e-01
## `PCIAT-PCIAT_18`4                        -1.131664e+00
## `PCIAT-PCIAT_18`5                        -1.072562e+00
## `PCIAT-PCIAT_18`Missing                   9.848147e-02
## `PCIAT-PCIAT_19...21`1                    8.869594e-02
## `PCIAT-PCIAT_19...21`2                    1.392434e-01
## `PCIAT-PCIAT_19...21`3                    1.529762e-03
## `PCIAT-PCIAT_19...21`4                   -4.116822e-01
## `PCIAT-PCIAT_19...21`5                   -1.778151e-02
## `PCIAT-PCIAT_19...21`Missing              4.493055e-01
## `PCIAT-PCIAT_20...22`1                    1.504391e-01
## `PCIAT-PCIAT_20...22`2                   -4.462599e-02
## `PCIAT-PCIAT_20...22`3                   -6.814589e-01
## `PCIAT-PCIAT_20...22`4                   -5.541143e-01
## `PCIAT-PCIAT_20...22`5                    2.179479e+00
## `PCIAT-PCIAT_20...22`Missing             -2.815233e-01
## `PCIAT-PCIAT_Total`1                      3.053949e-01
## `PCIAT-PCIAT_Total`10                     5.415050e-01
## `PCIAT-PCIAT_Total`11                     1.799721e-01
## `PCIAT-PCIAT_Total`12                     4.198790e-01
## `PCIAT-PCIAT_Total`13                     2.579820e-01
## `PCIAT-PCIAT_Total`14                     2.846201e-01
## `PCIAT-PCIAT_Total`15                     5.503873e-01
## `PCIAT-PCIAT_Total`16                     1.305753e-01
## `PCIAT-PCIAT_Total`17                     1.438736e-01
## `PCIAT-PCIAT_Total`18                     4.129104e-04
## `PCIAT-PCIAT_Total`19                     6.888562e-02
## `PCIAT-PCIAT_Total`2                      3.515732e-01
## `PCIAT-PCIAT_Total`20                     6.622593e-02
## `PCIAT-PCIAT_Total`21                    -9.420960e-02
## `PCIAT-PCIAT_Total`22                     3.496920e-02
## `PCIAT-PCIAT_Total`23                    -1.828882e-01
## `PCIAT-PCIAT_Total`24                    -4.793701e-01
## `PCIAT-PCIAT_Total`25                    -3.751107e-01
## `PCIAT-PCIAT_Total`26                    -6.218660e-01
## `PCIAT-PCIAT_Total`27                    -6.378234e-01
## `PCIAT-PCIAT_Total`28                    -7.552456e-01
## `PCIAT-PCIAT_Total`29                    -6.388633e-01
## `PCIAT-PCIAT_Total`3                      4.698689e-01
## `PCIAT-PCIAT_Total`30                    -9.085514e-01
## `PCIAT-PCIAT_Total`31                    -2.672046e-01
## `PCIAT-PCIAT_Total`32                    -2.834746e-01
## `PCIAT-PCIAT_Total`33                    -5.527491e-01
## `PCIAT-PCIAT_Total`34                    -4.771959e-01
## `PCIAT-PCIAT_Total`35                    -1.897202e-01
## `PCIAT-PCIAT_Total`36                    -1.773418e-01
## `PCIAT-PCIAT_Total`37                    -6.337067e-01
## `PCIAT-PCIAT_Total`38                    -4.345512e-01
## `PCIAT-PCIAT_Total`39                    -2.824865e-01
## `PCIAT-PCIAT_Total`4                      4.777177e-01
## `PCIAT-PCIAT_Total`40                     3.640841e-03
## `PCIAT-PCIAT_Total`41                    -2.355006e-01
## `PCIAT-PCIAT_Total`42                     7.961800e-02
```

```
## `PCIAT-PCIAT_Total`43                    7.707132e-01
## `PCIAT-PCIAT_Total`44                    5.190197e-01
## `PCIAT-PCIAT_Total`45                    4.704005e-01
## `PCIAT-PCIAT_Total`46                    6.082151e-01
## `PCIAT-PCIAT_Total`47                    4.983837e-01
## `PCIAT-PCIAT_Total`48                    1.214496e+00
## `PCIAT-PCIAT_Total`49                    1.264631e+00
## `PCIAT-PCIAT_Total`5                     7.415499e-01
## `PCIAT-PCIAT_Total`50                   -9.274777e-01
## `PCIAT-PCIAT_Total`51                   -1.237321e+00
## `PCIAT-PCIAT_Total`52                   -1.023454e+00
## `PCIAT-PCIAT_Total`53                   -7.042124e-01
## `PCIAT-PCIAT_Total`54                   -7.963860e-01
## `PCIAT-PCIAT_Total`55                   -3.422397e-01
## `PCIAT-PCIAT_Total`56                   -9.360617e-01
## `PCIAT-PCIAT_Total`57                   -1.182488e+00
## `PCIAT-PCIAT_Total`58                   -9.100352e-02
## `PCIAT-PCIAT_Total`59                   -5.790390e-01
## `PCIAT-PCIAT_Total`6                     7.839404e-01
## `PCIAT-PCIAT_Total`60                   -1.689254e-01
## `PCIAT-PCIAT_Total`61                    1.208085e+00
## `PCIAT-PCIAT_Total`62                    6.569143e-01
## `PCIAT-PCIAT_Total`63                    6.711004e-01
## `PCIAT-PCIAT_Total`64                    6.696309e-01
## `PCIAT-PCIAT_Total`65                    1.509000e+00
## `PCIAT-PCIAT_Total`66                    7.579159e-01
## `PCIAT-PCIAT_Total`67                    8.734494e-02
## `PCIAT-PCIAT_Total`68                    2.043771e+00
## `PCIAT-PCIAT_Total`69                    1.738536e+00
## `PCIAT-PCIAT_Total`7                     6.884834e-01
## `PCIAT-PCIAT_Total`70                    8.539920e-01
## `PCIAT-PCIAT_Total`71                    2.124345e+00
## `PCIAT-PCIAT_Total`72                    2.216970e+00
## `PCIAT-PCIAT_Total`73                   -1.560064e-01
## `PCIAT-PCIAT_Total`74                   -9.305289e-01
## `PCIAT-PCIAT_Total`75                   -5.625377e-01
## `PCIAT-PCIAT_Total`76                   -7.588897e-02
## `PCIAT-PCIAT_Total`77                    2.338510e+00
## `PCIAT-PCIAT_Total`78                    3.491732e-01
## `PCIAT-PCIAT_Total`79                   -1.845562e+00
## `PCIAT-PCIAT_Total`8                     8.702944e-01
## `PCIAT-PCIAT_Total`80                    3.567882e+00
## `PCIAT-PCIAT_Total`81                    3.252016e+00
## `PCIAT-PCIAT_Total`82                   -5.864072e+00
## `PCIAT-PCIAT_Total`83                    1.807663e+00
## `PCIAT-PCIAT_Total`84                   -2.438907e+00
## `PCIAT-PCIAT_Total`85                    1.578971e+00
## `PCIAT-PCIAT_Total`87                   -2.859654e+00
## `PCIAT-PCIAT_Total`88                    3.674699e-01
## `PCIAT-PCIAT_Total`89                   -3.591912e+00
## `PCIAT-PCIAT_Total`9                     8.254816e-01
## `PCIAT-PCIAT_Total`90                   -2.366295e+00
## `PCIAT-PCIAT_Total`91                    3.385090e+00
## `PCIAT-PCIAT_Total`92                   -1.756658e+00
```

```
## `PCIAT-PCIAT_Total`93                    -4.528152e+00
## `Basic_Demos-Age`                         2.768283e-03
## `Basic_Demos-Sex`                        -8.003231e-02
## `CGAS-CGAS_Score`                        -5.656962e-03
## `Physical-BMI`                            1.832432e-02
## `Physical-Waist_Circumference`            3.185923e-03
## `Physical-Diastolic_BP`                  -3.113819e-03
## `Physical-HeartRate`                     -1.488237e-03
## `Physical-Systolic_BP`                    4.649057e-03
## `Fitness_Endurance-Max_Stage`            -7.952457e-02
## `Fitness_Endurance-Time_Mins`             3.298288e-02
## `Fitness_Endurance-Time_Sec`              5.360093e-03
## `FGC-FGC_CU`                              3.018355e-03
## `FGC-FGC_CU_Zone`                        -1.835460e-02
## `FGC-FGC_GSND_Zone`                       1.772340e-02
## `FGC-FGC_GSD`                            -1.723896e-02
## `FGC-FGC_GSD_Zone`                        4.322701e-01
## `FGC-FGC_PU`                              7.117552e-03
## `FGC-FGC_PU_Zone`                        -8.066493e-02
## `FGC-FGC_SRL`                             3.877186e-03
## `FGC-FGC_SRL_Zone`                       -6.340450e-02
## `FGC-FGC_SRR_Zone`                        2.945818e-02
## `FGC-FGC_TL`                             -5.552148e-03
## `FGC-FGC_TL_Zone`                         1.603699e-01
## `BIA-BIA_Activity_Level_num`              1.318608e-03
## `BIA-BIA_BMI`                            -6.538817e-01
## `BIA-BIA_BMR`                            -8.069388e-07
## `BIA-BIA_FFMI`                            6.449409e-01
## `BIA-BIA_FMI`                             6.657636e-01
## `BIA-BIA_Frame_num`                      -4.993732e-02
## `PAQ_A-PAQ_A_Total`                       4.480376e-02
## `PAQ_C-PAQ_C_Total`                      -4.902305e-02
## `PCIAT-PCIAT_01...55`                    -4.034028e-02
## `PCIAT-PCIAT_02...56`                     7.051572e-02
## `PCIAT-PCIAT_03...57`                    -1.771159e-02
## `PCIAT-PCIAT_04...58`                    -3.200695e-02
## `PCIAT-PCIAT_05...59`                     1.338268e-01
## `PCIAT-PCIAT_06...60`                    -4.678912e-02
## `PCIAT-PCIAT_07...61`                     1.842800e-02
## `PCIAT-PCIAT_08...62`                     5.454250e-03
## `PCIAT-PCIAT_09...63`                     8.536891e-03
## `PCIAT-PCIAT_10...64`                     3.232957e-03
## `PCIAT-PCIAT_11...65`                    -1.219286e-03
## `PCIAT-PCIAT_12...66`                     7.443666e-02
## `PCIAT-PCIAT_13...67`                     2.776734e-02
## `PCIAT-PCIAT_14...68`                     1.637270e-03
## `PCIAT-PCIAT_15...69`                     1.006383e-02
## `PCIAT-PCIAT_16...70`                     3.352276e-02
## `PCIAT-PCIAT_17...71`                     9.190312e-02
## `PCIAT-PCIAT_19...72`                    -2.688196e-02
## `PCIAT-PCIAT_20...73`                     4.602723e-02
## `SDS-SDS_Total_T`                         4.458086e-03
## `PreInt_EduHx-computerinternet_hoursday` -3.869945e-02
##
```

```
## Proportion of trace:
##    LD1    LD2    LD3
## 0.9109 0.0722 0.0169
```

```r
# Predicting using the LDA model
lda_predictions = predict(lda_model_train, lda_data_train)

# Add predictions to the original data
lda_data_train$predicted_sii = lda_predictions$class

# Confusion matrix to evaluate classification accuracy
confusion_matrix = table(lda_data_train$sii, lda_data_train$predicted_sii)

print("Confusion Matrix:")
```

```
## [1] "Confusion Matrix:"
```

```r
print(confusion_matrix)
```

```
##
##        0    1    2    3
##   0 1558   36    0    0
##   1    0  730    0    0
##   2    0    0  378    0
##   3    0    0    0   34
```

```r
# Calculate accuracy
accuracy = sum(diag(confusion_matrix)) / sum(confusion_matrix)
print(paste("Accuracy:", round(accuracy * 100, 2), "%"))
```

```
## [1] "Accuracy: 98.68 %"
```

```r
error_rate = 1 - accuracy
print(paste("Error Rate:", round(error_rate * 100, 2), "%"))
```

```
## [1] "Error Rate: 1.32 %"
```

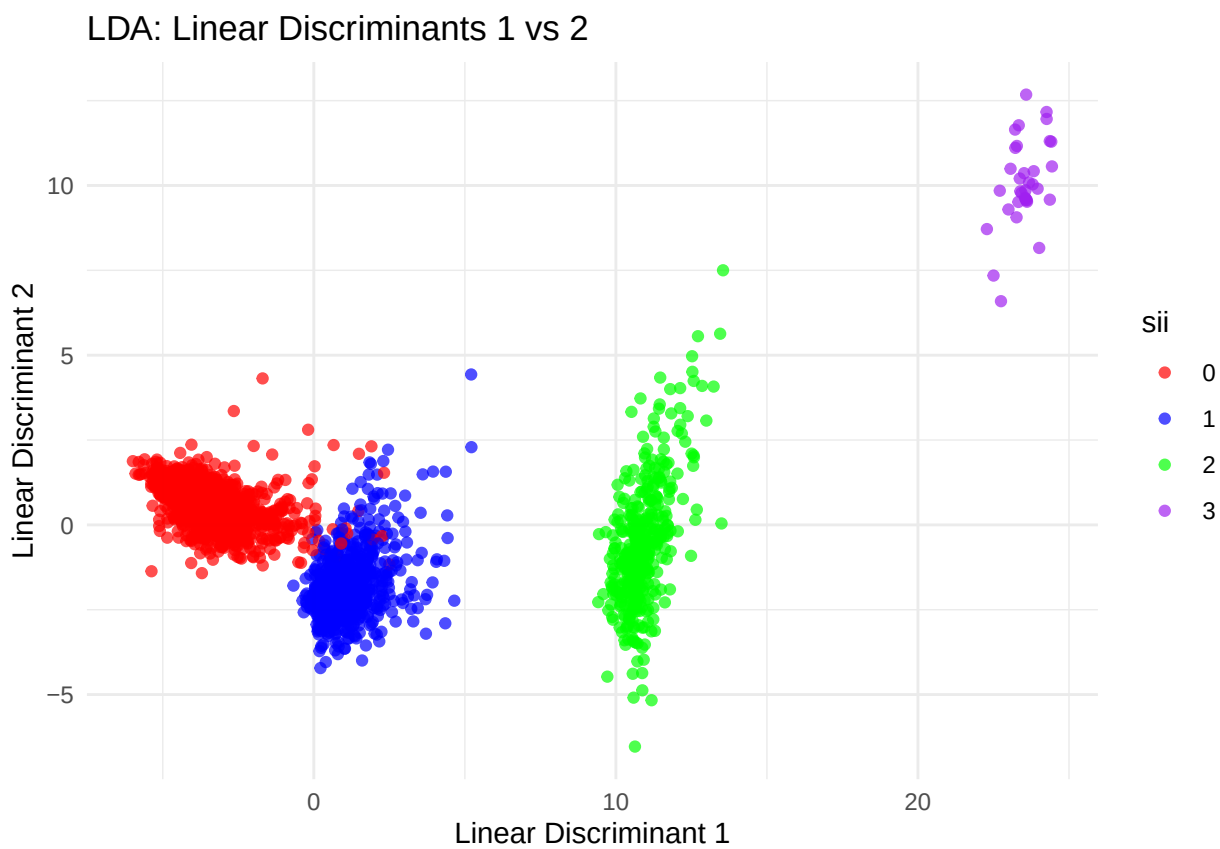
```r
# ----------------------------------------------------------------------------------------------
# LDA Visualization

# Check if the number of linear discriminants is sufficient
lda_discriminants = lda_predictions$x
lda_discriminants = as.data.frame(lda_discriminants)

if (ncol(lda_discriminants) >= 2) {
  # Extract the first two linear discriminants
  lda_data_train$LD1 = lda_discriminants[, 1]  # First Linear Discriminant
  lda_data_train$LD2 = lda_discriminants[, 2]  # Second Linear Discriminant
  # Plot the LDA results
  lda_plot_naRm = ggplot(lda_data_train, aes(x = LD1, y = LD2, color = sii)) +
    geom_point(alpha = 0.7) +
    labs(title = "LDA: Linear Discriminants 1 vs 2",
         x = "Linear Discriminant 1",
         y = "Linear Discriminant 2") +
    theme_minimal() +
    scale_color_manual(values = c("red", "blue", "green", "purple"))  # Adjust colors as needed
} else {
  print("Insufficient linear discriminants for visualization.")
```

```
}

print(lda_plot_naRm)
```




```
#ggsave("lda_plot_naRm", plot = lda_plot_naRm, width = 8, height = 6, dpi = 300)

# ----------------------------------------------------------------------------------------------------------------

# Testing
# Read the test dataset
test = read_csv("test.csv")
```

```
## Rows: 20 Columns: 59
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (11): id, Basic_Demos-Enroll_Season, CGAS-Season, Physical-Season, Fitne...
## dbl (48): Basic_Demos-Age, Basic_Demos-Sex, CGAS-CGAS_Score, Physical-BMI, P...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
colnames(test) = gsub("-", "_", colnames(test))

train_sii = subset(train_clean, select = c(id, sii))
test_clean = test %>% left_join(train_sii, by = "id", suffix = c("", "_train"))

# Remove NA
```

```r
test_clean = test_clean[!is.na(test_clean$sii), ]

test_clean$sii = factor(test_clean$sii, levels = levels(train_clean$sii))
#test_clean = test_clean[, !grepl("Season", names(test_clean))]

# Separate categorical and quantitative variables
# Select categorical variables (factor or character)
categorical_vars_test = test_clean[, sapply(test_clean, function(x) is.factor(x) | is.character(x))]

# Select quantitative variables (numeric)
quantitative_vars_test = test_clean[, sapply(test_clean, is.numeric)]

# Combine quantitative and categorical variables before imputation
test_combined = bind_cols(categorical_vars_test, quantitative_vars_test)
test_combined = test_combined[, !(names(test_combined) %in% c("PAQ_A_Season", "PAQ_A_PAQ_A_Total"))]


# Perform kNN imputation (with k = 3)
test_combined_imputed = kNN(test_combined, k = 3)

# Add random noise to numeric columns to maintain variability
test_combined_imputed = test_combined_imputed %>%
  mutate(across(where(is.numeric), ~ . + rnorm(length(.), mean = 0, sd = 0.01)))  # Add small noise

# Split back into quantitative variables (numeric) after imputation
quantitative_vars_test_imputed = test_combined_imputed[, sapply(test_combined_imputed, is.numeric)]
# Split back into categorical variables (factor or character) after imputation
categorical_vars_test_imputed = test_combined_imputed[, sapply(test_combined_imputed, function(x) is.cha

# Re-combine the imputed data
# This is necessary because impuation method add logical variables into dataset
test_data = bind_cols(categorical_vars_test_imputed, quantitative_vars_test_imputed)
test_data$sii = test_clean$sii  # Add 'sii' to the dataset (raw column)

lda_data_test = test_data
lda_data_test$sii = as.factor(lda_data_test$sii)

# Remove 'id' column or any non-predictor columns, if necessary
lda_data_test = subset(lda_data_test, select = -id)

# Perform Linear Discriminant Analysis
lda_model_test = lda(sii ~ ., data = lda_data_test)
```

```
## Warning in lda.default(x, grouping, ...): group 3 is empty

## Warning in lda.default(x, grouping, ...): variables are collinear
```

```r
print(lda_model_test)
```

```
## Call:
## lda(sii ~ ., data = lda_data_test)
##
## Prior probabilities of groups:
##         0         1         2
## 0.5454545 0.2727273 0.1818182
```

```
##
## Group means:
##   Basic_Demos_Enroll_SeasonSpring Basic_Demos_Enroll_SeasonSummer
## 0                       0.1666667                       0.3333333
## 1                       0.3333333                       0.0000000
## 2                       0.0000000                       0.0000000
##   Basic_Demos_Enroll_SeasonWinter CGAS_SeasonSpring CGAS_SeasonSummer
## 0                       0.1666667         0.1666667         0.3333333
## 1                       0.3333333         0.3333333         0.0000000
## 2                       0.5000000         0.0000000         0.5000000
##   CGAS_SeasonWinter Physical_SeasonSpring Physical_SeasonSummer
## 0         0.1666667             0.3333333             0.0000000
## 1         0.3333333             0.0000000             0.6666667
## 2         0.5000000             0.5000000             0.0000000
##   Physical_SeasonWinter Fitness_Endurance_SeasonSpring
## 0             0.1666667                       0.6666667
## 1             0.3333333                       0.3333333
## 2             0.0000000                       1.0000000
##   Fitness_Endurance_SeasonSummer FGC_SeasonSpring FGC_SeasonSummer
## 0                      0.0000000        0.1666667        0.0000000
## 1                      0.6666667        0.0000000        0.6666667
## 2                      0.0000000        0.5000000        0.0000000
##   FGC_SeasonWinter BIA_SeasonSummer BIA_SeasonWinter PAQ_C_SeasonSpring
## 0        0.3333333        0.5000000        0.3333333          0.0000000
## 1        0.3333333        0.6666667        0.3333333          0.3333333
## 2        0.0000000        0.5000000        0.0000000          0.0000000
##   PAQ_C_SeasonSummer PAQ_C_SeasonWinter SDS_SeasonSpring SDS_SeasonSummer
## 0          0.1666667          0.6666667        0.1666667        0.0000000
## 1          0.0000000          0.3333333        0.0000000        0.6666667
## 2          0.0000000          1.0000000        0.5000000        0.0000000
##   SDS_SeasonWinter PreInt_EduHx_SeasonSpring PreInt_EduHx_SeasonSummer
## 0              0.5                 0.1666667                 0.3333333
## 1              0.0                 0.3333333                 0.0000000
## 2              0.5                 0.0000000                 0.0000000
##   PreInt_EduHx_SeasonWinter Basic_Demos_Age Basic_Demos_Sex CGAS_CGAS_Score
## 0                 0.1666667        8.499019     0.336418688        66.16439
## 1                 0.3333333       11.665450     0.339380121        60.33226
## 2                 0.5000000        5.992990    -0.003555525        51.00087
##   Physical_BMI Physical_Height Physical_Weight Physical_Waist_Circumference
## 0     19.16975        49.16323        64.93443                     22.32922
## 1     20.55593        57.75531        97.93994                     21.99587
## 2     23.09634        49.99360        86.19167                     22.00392
##   Physical_Diastolic_BP Physical_HeartRate Physical_Systolic_BP
## 0              74.99758           81.66640             119.1611
## 1              60.99640           83.00314             123.0011
## 2              77.49059           79.00271             110.5018
##   Fitness_Endurance_Max_Stage Fitness_Endurance_Time_Mins
## 0                    5.003458                    7.003005
## 1                    5.314132                    7.671777
## 2                    4.492006                    6.000387
##   Fitness_Endurance_Time_Sec FGC_FGC_CU FGC_FGC_CU_Zone FGC_FGC_GSND
## 0                   33.00061   7.326973       0.6670757     15.44509
## 1                   34.33184  17.998285       0.6612572     17.40039
## 2                   32.49205   3.008467       0.4920480     16.49367
```

```
##   FGC_FGC_GSND_Zone FGC_FGC_GSD FGC_FGC_GSD_Zone   FGC_FGC_PU FGC_FGC_PU_Zone
## 0         1.0017617    17.37138         1.999403  2.333035955      0.16464161
## 1         1.3416918    18.06358         1.669512 10.335295459      0.33573401
## 2         0.9971492    17.89405         2.002898  0.003223332      0.01780804
##   FGC_FGC_SRL FGC_FGC_SRL_Zone FGC_FGC_SRR FGC_FGC_SRR_Zone FGC_FGC_TL
## 0    9.501123        0.8361002    9.334303        0.8292222   6.669608
## 1    8.326844        0.6589469    9.178169        0.6617080   9.174902
## 2    9.498861        0.5096528   10.502176        0.5046393   8.999811
##   FGC_FGC_TL_Zone BIA_BIA_Activity_Level_num BIA_BIA_BMC BIA_BIA_BMI
## 0       0.6638668                   1.985140    3.683286    18.80267
## 1       0.6597868                   3.334802    4.421398    23.18135
## 2       0.9935722                   1.995319    3.496065    18.97370
##   BIA_BIA_BMR BIA_BIA_DEE BIA_BIA_ECW BIA_BIA_FFM BIA_BIA_FFMI BIA_BIA_FMI
## 0    1115.067    1789.499    15.62503    61.03899     14.61204    4.192881
## 1    1233.956    2298.191    21.92491    73.69886     15.47560    7.710540
## 2    1085.979    1707.708    12.28209    52.41614     14.24899    4.008180
##   BIA_BIA_Fat BIA_BIA_Frame_num BIA_BIA_ICW BIA_BIA_LDM BIA_BIA_LST BIA_BIA_SMM
## 0    17.47127          1.670122    28.37030    15.39779    55.57168    24.92141
## 1    37.50175          2.007048    33.12050    18.66391    69.26601    32.69742
## 2    15.27894          1.501614    27.67968    12.84808    49.18293    23.01513
##   BIA_BIA_TBW PAQ_C_PAQ_C_Total SDS_SDS_Total_Raw SDS_SDS_Total_T
## 0    43.91807          2.323725          38.16677        54.16059
## 1    55.04267          3.194046          34.66733        49.32979
## 2    39.95144          2.260529          36.99407        52.51517
##   PreInt_EduHx_computerinternet_hoursday
## 0                              1.3326392
## 1                              0.3354493
## 2                              2.5155977
##
## Coefficients of linear discriminants:
##                                                LD1           LD2
## Basic_Demos_Enroll_SeasonSpring         1.108211e-01  0.1321220323
## Basic_Demos_Enroll_SeasonSummer         1.870732e-02  0.0588592821
## Basic_Demos_Enroll_SeasonWinter         1.397463e-01 -0.1971774740
## CGAS_SeasonSpring                       6.391270e-02  0.0221492838
## CGAS_SeasonSummer                       1.987209e-02  0.0248015055
## CGAS_SeasonWinter                       8.672412e-03  0.0725292976
## Physical_SeasonSpring                   1.987209e-02  0.0248015055
## Physical_SeasonSummer                  -1.401254e-02 -0.1785940682
## Physical_SeasonWinter                   6.391270e-02  0.0221492838
## Fitness_Endurance_SeasonSpring         -1.443115e-02 -0.0621067747
## Fitness_Endurance_SeasonSummer          5.878800e-03  0.0686017598
## FGC_SeasonSpring                        2.764040e-01  0.0065471993
## FGC_SeasonSummer                       -1.401254e-02 -0.1785940682
## FGC_SeasonWinter                       -1.181187e-01  0.0349818766
## BIA_SeasonSummer                       -6.285685e-02  0.0184802294
## BIA_SeasonWinter                        1.577604e-01 -0.0055854643
## PAQ_C_SeasonSpring                     -5.878800e-03 -0.0686017598
## PAQ_C_SeasonSummer                     -2.336505e-01  0.1474486765
## PAQ_C_SeasonWinter                     -1.518279e-02 -0.0759036241
## SDS_SeasonSpring                        2.764040e-01  0.0065471993
## SDS_SeasonSummer                       -1.401254e-02 -0.1785940682
## SDS_SeasonWinter                       -1.967409e-01 -0.0436043209
## PreInt_EduHx_SeasonSpring               1.108211e-01  0.1321220323
```

```
## PreInt_EduHx_SeasonSummer               1.870732e-02  0.0588592821
## PreInt_EduHx_SeasonWinter               1.397463e-01 -0.1971774740
## Basic_Demos_Age                        -3.156945e-02 -0.0233493783
## Basic_Demos_Sex                        -1.242451e-02  0.1574641424
## CGAS_CGAS_Score                        -1.372216e-03  0.0110923726
## Physical_BMI                            1.271174e-02 -0.0182099466
## Physical_Height                        -1.050924e-02 -0.0081356006
## Physical_Weight                         4.554921e-04 -0.0052641291
## Physical_Waist_Circumference            5.022935e-02 -0.0505201631
## Physical_Diastolic_BP                   1.755702e-03 -0.0060896820
## Physical_HeartRate                     -6.142170e-03  0.0001349717
## Physical_Systolic_BP                   -3.308331e-04 -0.0007449936
## Fitness_Endurance_Max_Stage            -1.760039e-01  0.1392316998
## Fitness_Endurance_Time_Mins            -8.676901e-02  0.0682738012
## Fitness_Endurance_Time_Sec             -1.937025e-02 -0.0053648574
## FGC_FGC_CU                             -9.047403e-03 -0.0026107889
## FGC_FGC_CU_Zone                        -7.284383e-02 -0.1562725167
## FGC_FGC_GSND                            3.301946e-02 -0.0120678066
## FGC_FGC_GSND_Zone                      -4.688601e-03 -0.0646728859
## FGC_FGC_GSD                             7.230412e-02 -0.0372638089
## FGC_FGC_GSD_Zone                       -1.675682e-02 -0.1825464503
## FGC_FGC_PU                             -2.008750e-04  0.0098597196
## FGC_FGC_PU_Zone                        -1.223546e-01  0.1611693943
## FGC_FGC_SRL                             1.207404e-02 -0.0156607025
## FGC_FGC_SRL_Zone                        5.738176e-02 -0.0393545121
## FGC_FGC_SRR                             1.840559e-02 -0.0214338519
## FGC_FGC_SRR_Zone                        5.495278e-02 -0.0415966802
## FGC_FGC_TL                              1.350605e-02 -0.0278472279
## FGC_FGC_TL_Zone                        -1.252802e-02 -0.0141808049
## BIA_BIA_Activity_Level_num              6.012490e-03  0.0614439972
## BIA_BIA_BMC                            -3.786028e-02  0.0175463181
## BIA_BIA_BMI                            -1.123148e-02 -0.0041636778
## BIA_BIA_BMR                            -3.400292e-04 -0.0002211358
## BIA_BIA_DEE                            -7.434592e-05  0.0001100450
## BIA_BIA_ECW                            -9.741096e-03 -0.0033084748
## BIA_BIA_FFM                            -5.568406e-03 -0.0005890834
## BIA_BIA_FFMI                           -8.298223e-02  0.0172221301
## BIA_BIA_FMI                            -1.289474e-02 -0.0049895037
## BIA_BIA_Fat                            -2.123454e-03 -0.0013551488
## BIA_BIA_Frame_num                      -6.088459e-02 -0.0583744414
## BIA_BIA_ICW                            -1.394307e-02  0.0010905631
## BIA_BIA_LDM                             2.892818e-03 -0.0138530122
## BIA_BIA_LST                            -4.655239e-03 -0.0021802226
## BIA_BIA_SMM                            -1.431957e-02  0.0017607612
## BIA_BIA_TBW                            -6.960168e-03 -0.0014299268
## PAQ_C_PAQ_C_Total                       9.529354e-02 -0.0925872751
## SDS_SDS_Total_Raw                      -3.999437e-03  0.0126587588
## SDS_SDS_Total_T                        -3.134131e-03  0.0102650060
## PreInt_EduHx_computerinternet_hoursday -1.360443e-02 -0.0378920496
##
## Proportion of trace:
##    LD1    LD2
## 0.7137 0.2863
```

```r
lda_predictions = predict(lda_model_test, lda_data_test)

# Add predictions to the original data
lda_data_test$predicted_sii = lda_predictions$class

# Confusion matrix:
confusion_matrix = table(lda_data_test$sii, lda_data_test$predicted_sii)

print("Confusion Matrix:")
```

```
## [1] "Confusion Matrix:"
```

```r
print(confusion_matrix)
```

```
##
##     0 1 2 3
##   0 6 0 0 0
##   1 2 1 0 0
##   2 1 0 1 0
##   3 0 0 0 0
```

```r
# Calculate accuracy
accuracy = sum(diag(confusion_matrix)) / sum(confusion_matrix)
print(paste("Accuracy:", round(accuracy * 100, 2), "%"))
```

```
## [1] "Accuracy: 72.73 %"
```

```r
error_rate = 1 - accuracy
print(paste("Error Rate:", round(error_rate * 100, 2), "%"))
```

```
## [1] "Error Rate: 27.27 %"
```

```r
# ----------------------------------------------------------------------------------------------------
```

Splitting the Training Data

```r
library(recipes)
library(readr)
library(dplyr)
library(caret)
library(ggplot2)
library(parsnip)
library(purrr)
library(discrim)
library(tidyr)
library(lda)
library(VIM)
library(MASS)
set.seed(445)
train = read_csv("train.csv")
```

```
## Rows: 3960 Columns: 82
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (12): id, Basic_Demos-Enroll_Season, CGAS-Season, Physical-Season, Fitne...
## dbl (70): Basic_Demos-Age, Basic_Demos-Sex, CGAS-CGAS_Score, Physical-BMI, P...
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
#train_clean = train[!is.na(train$sii), ]

colnames(train) = gsub("-", "_", colnames(train)) # Replacing hyphens because R can't handle those
#train_clean = train_clean[, !grepl("Season", names(train))]
train_clean$sii = factor(train_clean$sii) # Making sure `sii` is a factor with 4 levels (plus extra lev

categorical_vars = train_clean[, sapply(train_clean, function(x) is.factor(x) | is.character(x))]
# Select categorical variables including 'sii'
# Identify columns that contain 'PCIAT' in their names but excluding 'PCIAT_PCIAT_Total' (quant. variab
pciat_columns = grep("PCIAT", names(train_clean), value = TRUE)
pciat_columns = pciat_columns[pciat_columns != "PCIAT_PCIAT_Total"]
categorical_vars = cbind(categorical_vars, train_clean[, pciat_columns])

categorical_vars = categorical_vars %>%
  mutate(across(everything(), ~ replace(., is.na(.), "Missing")))
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `across(everything(), ~replace(., is.na(.), "Missing"))`.
## Caused by warning in `[<-.factor`:
## ! invalid factor level, NA generated
```

```r
quantitative_vars = train_clean[, sapply(train_clean, is.numeric)]

quantitative_vars_imputed = quantitative_vars %>%
  mutate(across(where(is.numeric), ~ ifelse(is.na(.), median(., na.rm = TRUE), .)))

# ---------------------------------------------------------------------------------------------------
# Removing collinear variables:

# Compute correlation matrix for quantitative variables
correlation_matrix = cor(quantitative_vars_imputed)

# Find highly correlated pairs
high_corr_pairs = findCorrelation(correlation_matrix, cutoff = 0.8, verbose = TRUE)
```

```
## Compare row 66  and column  48 with corr  0.822
##   Means:  0.287 vs 0.141 so flagging column 66
## Compare row 5  and column  6 with corr  0.809
##   Means:  0.217 vs 0.137 so flagging column 5
## Compare row 6  and column  4 with corr  0.845
##   Means:  0.202 vs 0.135 so flagging column 6
## Compare row 63  and column  61 with corr  0.838
##   Means:  0.2 vs 0.133 so flagging column 63
## Compare row 41  and column  42 with corr  0.994
##   Means:  0.195 vs 0.131 so flagging column 41
## Compare row 42  and column  39 with corr  0.997
##   Means:  0.174 vs 0.129 so flagging column 42
## Compare row 39  and column  32 with corr  0.993
##   Means:  0.16 vs 0.128 so flagging column 39
## Compare row 32  and column  43 with corr  0.994
##   Means:  0.147 vs 0.127 so flagging column 32
```

```
## Compare row 43  and column  31 with corr  1
##   Means:  0.128 vs 0.127 so flagging column 43
## Compare row 31  and column  34 with corr  1
##   Means:  0.11 vs 0.127 so flagging column 34
## Compare row 31  and column  33 with corr  0.999
##   Means:  0.095 vs 0.128 so flagging column 33
## Compare row 31  and column  37 with corr  0.979
##   Means:  0.079 vs 0.129 so flagging column 37
## Compare row 31  and column  40 with corr  0.999
##   Means:  0.063 vs 0.131 so flagging column 40
## Compare row 31  and column  29 with corr  0.99
##   Means:  0.046 vs 0.134 so flagging column 29
## Compare row 18  and column  16 with corr  0.867
##   Means:  0.127 vs 0.137 so flagging column 16
## Compare row 22  and column  24 with corr  0.914
##   Means:  0.106 vs 0.137 so flagging column 24
## Compare row 68  and column  67 with corr  0.996
##   Means:  0.094 vs 0.139 so flagging column 67
## All correlations <= 0.8
```

```r
# Remove highly correlated variables
quantitative_vars_filtered = quantitative_vars_imputed[, -high_corr_pairs]

train_data = bind_cols(categorical_vars, quantitative_vars_filtered)
```

```
## New names:
## * `PCIAT-PCIAT_01` -> `PCIAT-PCIAT_01...3`
## * `PCIAT-PCIAT_02` -> `PCIAT-PCIAT_02...4`
## * `PCIAT-PCIAT_03` -> `PCIAT-PCIAT_03...5`
## * `PCIAT-PCIAT_04` -> `PCIAT-PCIAT_04...6`
## * `PCIAT-PCIAT_05` -> `PCIAT-PCIAT_05...7`
## * `PCIAT-PCIAT_06` -> `PCIAT-PCIAT_06...8`
## * `PCIAT-PCIAT_07` -> `PCIAT-PCIAT_07...9`
## * `PCIAT-PCIAT_08` -> `PCIAT-PCIAT_08...10`
## * `PCIAT-PCIAT_09` -> `PCIAT-PCIAT_09...11`
## * `PCIAT-PCIAT_10` -> `PCIAT-PCIAT_10...12`
## * `PCIAT-PCIAT_11` -> `PCIAT-PCIAT_11...13`
## * `PCIAT-PCIAT_12` -> `PCIAT-PCIAT_12...14`
## * `PCIAT-PCIAT_13` -> `PCIAT-PCIAT_13...15`
## * `PCIAT-PCIAT_14` -> `PCIAT-PCIAT_14...16`
## * `PCIAT-PCIAT_15` -> `PCIAT-PCIAT_15...17`
## * `PCIAT-PCIAT_16` -> `PCIAT-PCIAT_16...18`
## * `PCIAT-PCIAT_17` -> `PCIAT-PCIAT_17...19`
## * `PCIAT-PCIAT_19` -> `PCIAT-PCIAT_19...21`
## * `PCIAT-PCIAT_20` -> `PCIAT-PCIAT_20...22`
## * `PCIAT-PCIAT_01` -> `PCIAT-PCIAT_01...55`
## * `PCIAT-PCIAT_02` -> `PCIAT-PCIAT_02...56`
## * `PCIAT-PCIAT_03` -> `PCIAT-PCIAT_03...57`
## * `PCIAT-PCIAT_04` -> `PCIAT-PCIAT_04...58`
## * `PCIAT-PCIAT_05` -> `PCIAT-PCIAT_05...59`
## * `PCIAT-PCIAT_06` -> `PCIAT-PCIAT_06...60`
## * `PCIAT-PCIAT_07` -> `PCIAT-PCIAT_07...61`
## * `PCIAT-PCIAT_08` -> `PCIAT-PCIAT_08...62`
## * `PCIAT-PCIAT_09` -> `PCIAT-PCIAT_09...63`
## * `PCIAT-PCIAT_10` -> `PCIAT-PCIAT_10...64`
```

```
## * `PCIAT-PCIAT_11` -> `PCIAT-PCIAT_11...65`
## * `PCIAT-PCIAT_12` -> `PCIAT-PCIAT_12...66`
## * `PCIAT-PCIAT_13` -> `PCIAT-PCIAT_13...67`
## * `PCIAT-PCIAT_14` -> `PCIAT-PCIAT_14...68`
## * `PCIAT-PCIAT_15` -> `PCIAT-PCIAT_15...69`
## * `PCIAT-PCIAT_16` -> `PCIAT-PCIAT_16...70`
## * `PCIAT-PCIAT_17` -> `PCIAT-PCIAT_17...71`
## * `PCIAT-PCIAT_19` -> `PCIAT-PCIAT_19...72`
## * `PCIAT-PCIAT_20` -> `PCIAT-PCIAT_20...73`
```

```r
quantitative_vars_filtered = quantitative_vars_filtered %>% mutate(sii = train_data$sii)

# ---------------------------------------------------------------------------------------------------
# Split the data into training and test sets (80% train, 20% test)

set.seed(445)  # Ensure reproducibility
trainIndex <- createDataPartition(train_data$sii, p = 0.8, list = FALSE)
train_set <- train_data[trainIndex, ]
test_set <- train_data[-trainIndex, ]

# ---------------------------------------------------------------------------------------------------
# Linear Discriminant Analysis (LDA)

lda_data_train = train_set
lda_data_train$sii = as.factor(lda_data_train$sii)  # Ensure `sii` is a factor for LDA

lda_data_train = subset(lda_data_train, select = -id)

# Perform Linear Discriminant Analysis
lda_model_train = lda(sii ~ ., data = lda_data_train)
```

```
## Warning in lda.default(x, grouping, ...): variables are collinear
```

```r
# View the LDA model output
print(lda_model_train)
```

```
## Call:
## lda(sii ~ ., data = lda_data_train)
##
## Prior probabilities of groups:
##          0          1          2          3
## 0.58238247 0.26654496 0.13829302 0.01277955
##
## Group means:
##   `PCIAT-PCIAT_01...3`1 `PCIAT-PCIAT_01...3`2 `PCIAT-PCIAT_01...3`3
## 0            0.17398119             0.2938871             0.1300940
## 1            0.05993151             0.1763699             0.2619863
## 2            0.01650165             0.0330033             0.1023102
## 3            0.03571429             0.0000000             0.0000000
##   `PCIAT-PCIAT_01...3`4 `PCIAT-PCIAT_01...3`5 `PCIAT-PCIAT_01...3`Missing
## 0            0.07445141            0.01802508                 0.001567398
## 1            0.30479452            0.15239726                 0.001712329
## 2            0.36303630            0.46864686                 0.000000000
## 3            0.21428571            0.75000000                 0.000000000
##   `PCIAT-PCIAT_02...4`1 `PCIAT-PCIAT_02...4`2 `PCIAT-PCIAT_02...4`3
```

```
## 0            0.18495298            0.26489028            0.09561129
## 1            0.05136986            0.20376712            0.29280822
## 2            0.01320132            0.02970297            0.08910891
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_02...4`4 `PCIAT-PCIAT_02...4`5 `PCIAT-PCIAT_02...4`Missing
## 0            0.04467085            0.00862069                 0.001567398
## 1            0.27054795            0.14383562                 0.000000000
## 2            0.40264026            0.45214521                 0.000000000
## 3            0.03571429            0.96428571                 0.000000000
##   `PCIAT-PCIAT_03...5`1 `PCIAT-PCIAT_03...5`2 `PCIAT-PCIAT_03...5`3
## 0            0.16614420            0.32601881             0.1402821
## 1            0.01541096            0.19178082             0.3339041
## 2            0.00000000            0.01980198             0.1023102
## 3            0.00000000            0.00000000             0.0000000
##   `PCIAT-PCIAT_03...5`4 `PCIAT-PCIAT_03...5`5 `PCIAT-PCIAT_03...5`Missing
## 0            0.04937304            0.01097179                 0.001567398
## 1            0.34589041            0.10616438                 0.003424658
## 2            0.47524752            0.39933993                 0.003300330
## 3            0.07142857            0.92857143                 0.000000000
##   `PCIAT-PCIAT_04...6`1 `PCIAT-PCIAT_04...6`2 `PCIAT-PCIAT_04...6`3
## 0            0.21238245            0.04388715           0.006269592
## 1            0.36986301            0.17465753           0.054794521
## 2            0.24422442            0.21452145           0.135313531
## 3            0.03571429            0.10714286           0.142857143
##   `PCIAT-PCIAT_04...6`4 `PCIAT-PCIAT_04...6`5 `PCIAT-PCIAT_04...6`Missing
## 0           0.002351097          0.0007836991                 0.001567398
## 1           0.049657534          0.0085616438                 0.001712329
## 2           0.178217822          0.0561056106                 0.003300330
## 3           0.250000000          0.4285714286                 0.000000000
##   `PCIAT-PCIAT_05...7`1 `PCIAT-PCIAT_05...7`2 `PCIAT-PCIAT_05...7`3
## 0            0.21943574            0.26175549             0.1089342
## 1            0.01883562            0.21061644             0.3047945
## 2            0.00330033            0.00990099             0.1056106
## 3            0.00000000            0.00000000             0.0000000
##   `PCIAT-PCIAT_05...7`4 `PCIAT-PCIAT_05...7`5 `PCIAT-PCIAT_05...7`Missing
## 0            0.04075235            0.01018809                 0.002351097
## 1            0.27054795            0.18493151                 0.003424658
## 2            0.33003300            0.54125413                 0.003300330
## 3            0.10714286            0.89285714                 0.000000000
##   `PCIAT-PCIAT_06...8`1 `PCIAT-PCIAT_06...8`2 `PCIAT-PCIAT_06...8`3
## 0            0.31818182            0.05485893           0.007836991
## 1            0.41095890            0.27054795           0.090753425
## 2            0.21452145            0.21782178           0.141914191
## 3            0.03571429            0.07142857           0.000000000
##   `PCIAT-PCIAT_06...8`4 `PCIAT-PCIAT_06...8`5 `PCIAT-PCIAT_06...8`Missing
## 0           0.003134796             0.0000000                 0.002351097
## 1           0.039383562             0.0119863                 0.000000000
## 2           0.237623762             0.0990099                 0.003300330
## 3           0.321428571             0.5714286                 0.000000000
##   `PCIAT-PCIAT_07...9`1 `PCIAT-PCIAT_07...9`2 `PCIAT-PCIAT_07...9`3
## 0             0.1536050            0.03291536           0.005485893
## 1             0.3030822            0.10616438           0.035958904
## 2             0.2805281            0.12211221           0.056105611
## 3             0.1071429            0.32142857           0.071428571
```

```
##   `PCIAT-PCIAT_07...9`4 `PCIAT-PCIAT_07...9`5 `PCIAT-PCIAT_07...9`Missing
## 0           0.001567398            0.00000000                 0.003134796
## 1           0.037671233            0.01027397                 0.001712329
## 2           0.112211221            0.03960396                 0.006600660
## 3           0.142857143            0.28571429                 0.000000000
##   `PCIAT-PCIAT_08...10`1 `PCIAT-PCIAT_08...10`2 `PCIAT-PCIAT_08...10`3
## 0              0.3087774             0.09404389             0.01410658
## 1              0.3390411             0.31678082             0.14041096
## 2              0.1155116             0.20462046             0.22112211
## 3              0.0000000             0.00000000             0.03571429
##   `PCIAT-PCIAT_08...10`4 `PCIAT-PCIAT_08...10`5 `PCIAT-PCIAT_08...10`Missing
## 0            0.002351097            0.000000000                  0.003134796
## 1            0.061643836            0.006849315                  0.003424658
## 2            0.323432343            0.089108911                  0.000000000
## 3            0.250000000            0.714285714                  0.000000000
##   `PCIAT-PCIAT_09...11`1 `PCIAT-PCIAT_09...11`2 `PCIAT-PCIAT_09...11`3
## 0             0.30642633             0.06191223            0.006269592
## 1             0.45034247             0.27226027            0.071917808
## 2             0.22772277             0.22112211            0.155115512
## 3             0.07142857             0.00000000            0.107142857
##   `PCIAT-PCIAT_09...11`4 `PCIAT-PCIAT_09...11`5 `PCIAT-PCIAT_09...11`Missing
## 0            0.002351097            0.001567398                  0.002351097
## 1            0.032534247            0.015410959                  0.005136986
## 2            0.198019802            0.132013201                  0.000000000
## 3            0.178571429            0.642857143                  0.000000000
##   `PCIAT-PCIAT_10...12`1 `PCIAT-PCIAT_10...12`2 `PCIAT-PCIAT_10...12`3
## 0             0.29310345              0.1340125             0.01018809
## 1             0.28424658              0.3989726             0.14726027
## 2             0.10231023              0.2277228             0.20792079
## 3             0.03571429              0.0000000             0.00000000
##   `PCIAT-PCIAT_10...12`4 `PCIAT-PCIAT_10...12`5 `PCIAT-PCIAT_10...12`Missing
## 0            0.007053292            0.000000000                  0.001567398
## 1            0.061643836            0.005136986                  0.001712329
## 2            0.320132013            0.095709571                  0.000000000
## 3            0.321428571            0.642857143                  0.000000000
##   `PCIAT-PCIAT_11...13`1 `PCIAT-PCIAT_11...13`2 `PCIAT-PCIAT_11...13`3
## 0             0.20924765              0.2225705             0.06739812
## 1             0.12671233              0.2739726             0.22602740
## 2             0.04950495              0.1056106             0.18151815
## 3             0.00000000              0.0000000             0.03571429
##   `PCIAT-PCIAT_11...13`4 `PCIAT-PCIAT_11...13`5 `PCIAT-PCIAT_11...13`Missing
## 0             0.02037618             0.00000000                 0.0007836991
## 1             0.19691781             0.04965753                 0.0000000000
## 2             0.42904290             0.16501650                 0.0033003300
## 3             0.39285714             0.39285714                 0.0000000000
##   `PCIAT-PCIAT_12...14`1 `PCIAT-PCIAT_12...14`2 `PCIAT-PCIAT_12...14`3
## 0             0.09796238             0.00000000            0.000000000
## 1             0.32534247             0.01369863            0.003424658
## 2             0.35643564             0.04620462            0.009900990
## 3             0.32142857             0.17857143            0.071428571
##   `PCIAT-PCIAT_12...14`4 `PCIAT-PCIAT_12...14`5 `PCIAT-PCIAT_12...14`Missing
## 0             0.00000000             0.00000000                 0.0007836991
## 1             0.00000000             0.00000000                 0.0034246575
## 2             0.01320132             0.00000000                 0.0066006601
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## 3             0.07142857             0.03571429                 0.0000000000
##   `PCIAT-PCIAT_13...15`1 `PCIAT-PCIAT_13...15`2 `PCIAT-PCIAT_13...15`3
## 0             0.26645768              0.1206897             0.01567398
## 1             0.33904110              0.3321918             0.14554795
## 2             0.08910891              0.2112211             0.21782178
## 3             0.00000000              0.1071429             0.00000000
##   `PCIAT-PCIAT_13...15`4 `PCIAT-PCIAT_13...15`5 `PCIAT-PCIAT_13...15`Missing
## 0            0.004702194           0.0007836991                  0.003918495
## 1            0.077054795           0.0171232877                  0.001712329
## 2            0.297029703           0.1650165017                  0.003300330
## 3            0.178571429           0.7142857143                  0.000000000
##   `PCIAT-PCIAT_14...16`1 `PCIAT-PCIAT_14...16`2 `PCIAT-PCIAT_14...16`3
## 0              0.2539185             0.05642633            0.004702194
## 1              0.3732877             0.24143836            0.131849315
## 2              0.1122112             0.21782178            0.188118812
## 3              0.0000000             0.07142857            0.107142857
##   `PCIAT-PCIAT_14...16`4 `PCIAT-PCIAT_14...16`5 `PCIAT-PCIAT_14...16`Missing
## 0            0.001567398            0.000000000                  0.001567398
## 1            0.035958904            0.003424658                  0.000000000
## 2            0.247524752            0.108910891                  0.000000000
## 3            0.178571429            0.642857143                  0.000000000
##   `PCIAT-PCIAT_15...17`1 `PCIAT-PCIAT_15...17`2 `PCIAT-PCIAT_15...17`3
## 0             0.28526646             0.12460815             0.02429467
## 1             0.20034247             0.36472603             0.23630137
## 2             0.02970297             0.11881188             0.26072607
## 3             0.00000000             0.03571429             0.03571429
##   `PCIAT-PCIAT_15...17`4 `PCIAT-PCIAT_15...17`5 `PCIAT-PCIAT_15...17`Missing
## 0            0.003918495            0.001567398                  0.003918495
## 1            0.099315068            0.030821918                  0.000000000
## 2            0.359735974            0.204620462                  0.003300330
## 3            0.178571429            0.750000000                  0.000000000
##   `PCIAT-PCIAT_16...18`1 `PCIAT-PCIAT_16...18`2 `PCIAT-PCIAT_16...18`3
## 0             0.28840125             0.14263323             0.02429467
## 1             0.31164384             0.31849315             0.15924658
## 2             0.07920792             0.13201320             0.13201320
## 3             0.03571429             0.03571429             0.03571429
##   `PCIAT-PCIAT_16...18`4 `PCIAT-PCIAT_16...18`5 `PCIAT-PCIAT_16...18`Missing
## 0            0.007053292           0.0007836991                  0.003134796
## 1            0.089041096           0.0308219178                  0.001712329
## 2            0.330033003           0.2871287129                  0.003300330
## 3            0.142857143           0.7500000000                  0.000000000
##   `PCIAT-PCIAT_17...19`1 `PCIAT-PCIAT_17...19`2 `PCIAT-PCIAT_17...19`3
## 0             0.28761755             0.18652038             0.03918495
## 1             0.16609589             0.41609589             0.26883562
## 2             0.05280528             0.09240924             0.24422442
## 3             0.00000000             0.00000000             0.00000000
##   `PCIAT-PCIAT_17...19`4 `PCIAT-PCIAT_17...19`5 `PCIAT-PCIAT_17...19`Missing
## 0            0.007053292            0.001567398                  0.002351097
## 1            0.094178082            0.015410959                  0.003424658
## 2            0.402640264            0.184818482                  0.013201320
## 3            0.214285714            0.750000000                  0.035714286
##   `PCIAT-PCIAT_18`1 `PCIAT-PCIAT_18`2 `PCIAT-PCIAT_18`3 `PCIAT-PCIAT_18`4
## 0        0.30721003        0.16849530        0.03056426       0.006269592
## 1        0.21575342        0.39212329        0.18664384       0.106164384
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## 2        0.03630363        0.10231023        0.17821782       0.330033003
## 3        0.00000000        0.03571429        0.00000000       0.071428571
##   `PCIAT-PCIAT_18`5 `PCIAT-PCIAT_18`Missing `PCIAT-PCIAT_19...21`1
## 0       0.003134796             0.001567398             0.30094044
## 1       0.039383562             0.003424658             0.36130137
## 2       0.333333333             0.003300330             0.15181518
## 3       0.892857143             0.000000000             0.03571429
##   `PCIAT-PCIAT_19...21`2 `PCIAT-PCIAT_19...21`3 `PCIAT-PCIAT_19...21`4
## 0             0.06739812            0.007053292            0.003134796
## 1             0.30650685            0.101027397            0.056506849
## 2             0.22112211            0.165016502            0.290429043
## 3             0.00000000            0.142857143            0.285714286
##   `PCIAT-PCIAT_19...21`5 `PCIAT-PCIAT_19...21`Missing `PCIAT-PCIAT_20...22`1
## 0            0.002351097                  0.002351097              0.2821317
## 1            0.025684932                  0.003424658              0.4897260
## 2            0.115511551                  0.003300330              0.1980198
## 3            0.535714286                  0.000000000              0.0000000
##   `PCIAT-PCIAT_20...22`2 `PCIAT-PCIAT_20...22`3 `PCIAT-PCIAT_20...22`4
## 0             0.03291536           0.0007836991             0.00000000
## 1             0.24143836           0.0530821918             0.02568493
## 2             0.28052805           0.1650165017             0.22112211
## 3             0.07142857           0.0714285714             0.25000000
##   `PCIAT-PCIAT_20...22`5 `PCIAT-PCIAT_20...22`Missing `PCIAT-PCIAT_Total`1
## 0           0.0007836991                 0.0007836991           0.02115987
## 1           0.0017123288                 0.0017123288           0.00000000
## 2           0.0660066007                 0.0000000000           0.00000000
## 3           0.6071428571                 0.0000000000           0.00000000
##   `PCIAT-PCIAT_Total`10 `PCIAT-PCIAT_Total`11 `PCIAT-PCIAT_Total`12
## 0            0.02586207            0.01880878            0.01802508
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`13 `PCIAT-PCIAT_Total`14 `PCIAT-PCIAT_Total`15
## 0            0.01880878            0.03134796            0.01802508
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`16 `PCIAT-PCIAT_Total`17 `PCIAT-PCIAT_Total`18
## 0            0.02742947            0.02742947            0.03134796
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`19 `PCIAT-PCIAT_Total`2 `PCIAT-PCIAT_Total`20
## 0            0.02586207           0.03056426            0.04075235
## 1            0.00000000           0.00000000            0.00000000
## 2            0.00000000           0.00000000            0.00000000
## 3            0.00000000           0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`21 `PCIAT-PCIAT_Total`22 `PCIAT-PCIAT_Total`23
## 0            0.03605016            0.02664577            0.03840125
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`24 `PCIAT-PCIAT_Total`25 `PCIAT-PCIAT_Total`26
## 0            0.03056426            0.03134796            0.03996865
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## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`27 `PCIAT-PCIAT_Total`28 `PCIAT-PCIAT_Total`29
## 0            0.04231975            0.02742947            0.02742947
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`3 `PCIAT-PCIAT_Total`30 `PCIAT-PCIAT_Total`31
## 0           0.01724138            0.03918495            0.00000000
## 1           0.00000000            0.00000000            0.08732877
## 2           0.00000000            0.00000000            0.00000000
## 3           0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`32 `PCIAT-PCIAT_Total`33 `PCIAT-PCIAT_Total`34
## 0            0.00000000            0.00000000            0.00000000
## 1            0.06849315            0.05650685            0.06849315
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`35 `PCIAT-PCIAT_Total`36 `PCIAT-PCIAT_Total`37
## 0            0.00000000            0.00000000            0.00000000
## 1            0.05821918            0.07363014            0.05650685
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`38 `PCIAT-PCIAT_Total`39 `PCIAT-PCIAT_Total`4
## 0            0.00000000            0.00000000           0.01645768
## 1            0.03424658            0.05479452           0.00000000
## 2            0.00000000            0.00000000           0.00000000
## 3            0.00000000            0.00000000           0.00000000
##   `PCIAT-PCIAT_Total`40 `PCIAT-PCIAT_Total`41 `PCIAT-PCIAT_Total`42
## 0            0.00000000            0.00000000            0.00000000
## 1            0.05650685            0.04623288            0.06335616
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`43 `PCIAT-PCIAT_Total`44 `PCIAT-PCIAT_Total`45
## 0            0.00000000            0.00000000            0.00000000
## 1            0.04794521            0.05136986            0.04452055
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`46 `PCIAT-PCIAT_Total`47 `PCIAT-PCIAT_Total`48
## 0            0.00000000            0.00000000            0.00000000
## 1            0.02910959            0.04109589            0.02739726
## 2            0.00000000            0.00000000            0.00000000
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`49 `PCIAT-PCIAT_Total`5 `PCIAT-PCIAT_Total`50
## 0            0.00000000           0.01489028            0.00000000
## 1            0.03424658           0.00000000            0.00000000
## 2            0.00000000           0.00000000            0.05940594
## 3            0.00000000           0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`51 `PCIAT-PCIAT_Total`52 `PCIAT-PCIAT_Total`53
## 0             0.0000000            0.00000000             0.0000000
## 1             0.0000000            0.00000000             0.0000000
## 2             0.0660066            0.01650165             0.0660066
## 3             0.0000000            0.00000000             0.0000000
##   `PCIAT-PCIAT_Total`54 `PCIAT-PCIAT_Total`55 `PCIAT-PCIAT_Total`56
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## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.07590759            0.04290429            0.03630363
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`57 `PCIAT-PCIAT_Total`58 `PCIAT-PCIAT_Total`59
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.04290429            0.05610561            0.05280528
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`6 `PCIAT-PCIAT_Total`60 `PCIAT-PCIAT_Total`61
## 0           0.01724138             0.0000000            0.00000000
## 1           0.00000000             0.0000000            0.00000000
## 2           0.00000000             0.0330033            0.04620462
## 3           0.00000000             0.0000000            0.00000000
##   `PCIAT-PCIAT_Total`62 `PCIAT-PCIAT_Total`63 `PCIAT-PCIAT_Total`64
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.03960396            0.01650165            0.04620462
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`65 `PCIAT-PCIAT_Total`66 `PCIAT-PCIAT_Total`67
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.02640264            0.03630363            0.02640264
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`68 `PCIAT-PCIAT_Total`69 `PCIAT-PCIAT_Total`7
## 0            0.00000000            0.00000000           0.02664577
## 1            0.00000000            0.00000000           0.00000000
## 2            0.02640264            0.01320132           0.00000000
## 3            0.00000000            0.00000000           0.00000000
##   `PCIAT-PCIAT_Total`70 `PCIAT-PCIAT_Total`71 `PCIAT-PCIAT_Total`72
## 0             0.0000000            0.00000000            0.00000000
## 1             0.0000000            0.00000000            0.00000000
## 2             0.0330033            0.01650165            0.02640264
## 3             0.0000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`73 `PCIAT-PCIAT_Total`74 `PCIAT-PCIAT_Total`75
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.01320132            0.02970297            0.01650165
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`76 `PCIAT-PCIAT_Total`77 `PCIAT-PCIAT_Total`78
## 0            0.00000000            0.00000000            0.00000000
## 1            0.00000000            0.00000000            0.00000000
## 2            0.00990099            0.01980198            0.00990099
## 3            0.00000000            0.00000000            0.00000000
##   `PCIAT-PCIAT_Total`8 `PCIAT-PCIAT_Total`80 `PCIAT-PCIAT_Total`81
## 0           0.01645768             0.0000000             0.0000000
## 1           0.00000000             0.0000000             0.0000000
## 2           0.00000000             0.0000000             0.0000000
## 3           0.00000000             0.1071429             0.2142857
##   `PCIAT-PCIAT_Total`82 `PCIAT-PCIAT_Total`83 `PCIAT-PCIAT_Total`85
## 0            0.00000000             0.0000000            0.00000000
## 1            0.00000000             0.0000000            0.00000000
## 2            0.00000000             0.0000000            0.00000000
## 3            0.03571429             0.1071429            0.07142857
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##   `PCIAT-PCIAT_Total`87 `PCIAT-PCIAT_Total`88 `PCIAT-PCIAT_Total`89
## 0             0.0000000             0.0000000            0.00000000
## 1             0.0000000             0.0000000            0.00000000
## 2             0.0000000             0.0000000            0.00000000
## 3             0.1071429             0.1428571            0.03571429
##   `PCIAT-PCIAT_Total`9 `PCIAT-PCIAT_Total`90 `PCIAT-PCIAT_Total`91
## 0           0.01959248            0.00000000            0.00000000
## 1           0.00000000            0.00000000            0.00000000
## 2           0.00000000            0.00000000            0.00000000
## 3           0.00000000            0.03571429            0.03571429
##   `PCIAT-PCIAT_Total`92 `PCIAT-PCIAT_Total`93 `Basic_Demos-Age`
## 0            0.00000000            0.00000000           9.21395
## 1            0.00000000            0.00000000          11.11986
## 2            0.00000000            0.00000000          12.23762
## 3            0.07142857            0.03571429          14.42857
##   `Basic_Demos-Sex` `CGAS-CGAS_Score` `Physical-BMI`
## 0         0.4036050          65.67320       18.24415
## 1         0.3321918          64.64212       19.55753
## 2         0.2673267          63.99010       20.48007
## 3         0.1785714          60.75000       25.25036
##   `Physical-Waist_Circumference` `Physical-Diastolic_BP` `Physical-HeartRate`
## 0                       25.95141                68.91771             81.79937
## 1                       26.41952                70.48288             82.25171
## 2                       26.37294                70.12871             80.97030
## 3                       26.35714                71.28571             76.89286
##   `Physical-Systolic_BP` `Fitness_Endurance-Max_Stage`
## 0               114.7861                      5.053292
## 1               118.7466                      4.964041
## 2               119.3663                      4.976898
## 3               127.0000                      5.000000
##   `Fitness_Endurance-Time_Mins` `Fitness_Endurance-Time_Sec` `FGC-FGC_CU`
## 0                      7.193574                     27.63558     9.731975
## 1                      7.080479                     28.41952    13.369863
## 2                      7.023102                     27.75908    13.950495
## 3                      7.035714                     27.32143    18.464286
##   `FGC-FGC_CU_Zone` `FGC-FGC_GSND_Zone` `FGC-FGC_GSD` `FGC-FGC_GSD_Zone`
## 0         0.6724138            1.956897      21.04043           1.966301
## 1         0.6952055            1.955479      21.79384           1.984589
## 2         0.6303630            1.874587      23.12726           1.897690
## 3         0.7142857            1.964286      26.85000           2.035714
##   `FGC-FGC_PU` `FGC-FGC_PU_Zone` `FGC-FGC_SRL` `FGC-FGC_SRL_Zone`
## 0     4.359718         0.2374608      8.987187          0.7742947
## 1     5.758562         0.2602740      8.758733          0.7380137
## 2     5.953795         0.2244224      8.562541          0.6897690
## 3     9.178571         0.2500000      8.785714          0.5714286
##   `FGC-FGC_SRR_Zone` `FGC-FGC_TL` `FGC-FGC_TL_Zone`
## 0          0.7609718     8.892241         0.8393417
## 1          0.7568493     9.430137         0.8613014
## 2          0.7062706     9.263201         0.8283828
## 3          0.6428571     9.567857         0.8214286
##   `BIA-BIA_Activity_Level_num` `BIA-BIA_BMI` `BIA-BIA_BMR` `BIA-BIA_FFMI`
## 0                     2.764890      18.21236      1184.717       14.49461
## 1                     2.839041      19.00751      1195.817       14.57495
## 2                     2.815182      19.92005      1288.196       15.81330
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## 3                    3.142857      23.77579      1451.181       16.64169
##   `BIA-BIA_FMI` `BIA-BIA_Frame_num` `PAQ_A-PAQ_A_Total` `PAQ_C-PAQ_C_Total`
## 0      3.667797            1.764107            2.100312            2.580094
## 1      4.384334            1.880137            2.092019            2.583426
## 2      4.051968            1.953795            2.102320            2.587129
## 3      7.086459            2.214286            2.171393            2.528929
##   `PCIAT-PCIAT_01...55` `PCIAT-PCIAT_02...56` `PCIAT-PCIAT_03...57`
## 0              1.543103              1.226489              1.494514
## 1              3.183219              3.138699              3.321918
## 2              4.184818              4.211221              4.250825
## 3              4.642857              4.964286              4.928571
##   `PCIAT-PCIAT_04...58` `PCIAT-PCIAT_05...59` `PCIAT-PCIAT_06...60`
## 0             0.3322884              1.288401             0.4663009
## 1             1.1250000              3.368151             1.4417808
## 2             2.0726073              4.372937             2.5247525
## 3             3.8214286              4.892857             4.3214286
##   `PCIAT-PCIAT_07...61` `PCIAT-PCIAT_08...62` `PCIAT-PCIAT_09...63`
## 0             0.2421630             0.5517241              0.468652
## 1             0.8253425             1.6780822              1.422945
## 2             1.3399340             2.9273927              2.587459
## 3             2.9642857             4.6785714              4.321429
##   `PCIAT-PCIAT_10...64` `PCIAT-PCIAT_11...65` `PCIAT-PCIAT_12...66`
## 0             0.6214734             0.9396552            0.09796238
## 1             1.7979452             2.3886986            0.36301370
## 2             2.9405941             3.3531353            0.53135314
## 3             4.5357143             3.6428571            1.35714286
##   `PCIAT-PCIAT_13...67` `PCIAT-PCIAT_14...68` `PCIAT-PCIAT_15...69`
## 0             0.5815047             0.3887147             0.6347962
## 1             1.8356164             1.4126712             2.1900685
## 2             3.1815182             2.6468647             3.5148515
## 3             4.5000000             4.3928571             4.6428571
##   `PCIAT-PCIAT_16...70` `PCIAT-PCIAT_17...71` `PCIAT-PCIAT_19...72`
## 0             0.6818182             0.8166144             0.4835423
## 1             1.9383562             2.2619863             1.6352740
## 2             3.4983498             3.5181518             2.8316832
## 3             4.5357143             4.6428571             4.2857143
##   `PCIAT-PCIAT_20...73` `SDS-SDS_Total_T`
## 0             0.3550157          55.41850
## 1             1.2448630          58.65925
## 2             2.4686469          62.79868
## 3             4.3928571          66.71429
##   `PreInt_EduHx-computerinternet_hoursday`
## 0                                0.7374608
## 1                                1.2636986
## 2                                1.6303630
## 3                                1.9285714
##
## Coefficients of linear discriminants:
##                                                    LD1           LD2
## `PCIAT-PCIAT_01...3`1                    -6.551978e-01  2.570585e-01
## `PCIAT-PCIAT_01...3`2                    -4.665407e-01  3.557110e-01
## `PCIAT-PCIAT_01...3`3                    -2.730378e-01  4.839011e-02
## `PCIAT-PCIAT_01...3`4                     2.317290e-01 -1.068882e-01
## `PCIAT-PCIAT_01...3`5                     5.982687e-01 -1.163325e-01
```

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## `PCIAT-PCIAT_01...3`Missing              -2.597212e+00 -6.531465e+00
## `PCIAT-PCIAT_02...4`1                    -1.980572e-01  3.388893e-02
## `PCIAT-PCIAT_02...4`2                    -2.639939e-01  1.303349e-02
## `PCIAT-PCIAT_02...4`3                     5.602748e-02 -1.674658e-01
## `PCIAT-PCIAT_02...4`4                     2.051242e-01 -3.380400e-01
## `PCIAT-PCIAT_02...4`5                     6.948732e-01  1.424740e-01
## `PCIAT-PCIAT_02...4`Missing               1.113495e+01  1.224141e+01
## `PCIAT-PCIAT_03...5`1                    -1.457146e+00  5.771043e-01
## `PCIAT-PCIAT_03...5`2                    -1.236699e+00  5.332338e-01
## `PCIAT-PCIAT_03...5`3                    -7.009607e-01  1.243255e-01
## `PCIAT-PCIAT_03...5`4                     1.228538e-01 -2.078856e-01
## `PCIAT-PCIAT_03...5`5                     7.657082e-01 -5.942990e-01
## `PCIAT-PCIAT_03...5`Missing              -1.752520e+00 -1.179926e+00
## `PCIAT-PCIAT_04...6`1                     2.403798e-02 -2.010013e-01
## `PCIAT-PCIAT_04...6`2                     1.882130e-01 -2.082823e-01
## `PCIAT-PCIAT_04...6`3                     4.928465e-01  2.672129e-01
## `PCIAT-PCIAT_04...6`4                     6.672373e-01 -9.573653e-02
## `PCIAT-PCIAT_04...6`5                     1.319650e+00  1.912859e+00
## `PCIAT-PCIAT_04...6`Missing               3.598487e-01  9.767906e-01
## `PCIAT-PCIAT_05...7`1                    -7.588044e-01  6.294378e-01
## `PCIAT-PCIAT_05...7`2                    -6.233701e-01  3.443501e-01
## `PCIAT-PCIAT_05...7`3                    -2.777612e-01 -1.178521e-01
## `PCIAT-PCIAT_05...7`4                     1.026257e-01 -3.137478e-01
## `PCIAT-PCIAT_05...7`5                     8.772139e-01 -2.953725e-01
## `PCIAT-PCIAT_05...7`Missing              -1.548898e+00 -1.478622e+00
## `PCIAT-PCIAT_06...8`1                    -2.151094e-02 -2.197661e-01
## `PCIAT-PCIAT_06...8`2                     2.185961e-01 -3.351049e-01
## `PCIAT-PCIAT_06...8`3                     4.180482e-01 -3.783976e-01
## `PCIAT-PCIAT_06...8`4                     7.621191e-01  7.668890e-01
## `PCIAT-PCIAT_06...8`5                     9.437519e-01  1.385715e+00
## `PCIAT-PCIAT_06...8`Missing              -1.401507e-01  9.710271e-01
## `PCIAT-PCIAT_07...9`1                     1.194858e-01 -7.617591e-02
## `PCIAT-PCIAT_07...9`2                     5.086661e-01  8.927597e-02
## `PCIAT-PCIAT_07...9`3                     2.601537e-01  7.679577e-02
## `PCIAT-PCIAT_07...9`4                     7.004318e-01 -3.899734e-03
## `PCIAT-PCIAT_07...9`5                     9.357650e-01  9.703709e-01
## `PCIAT-PCIAT_07...9`Missing              -7.364964e-01 -1.696788e+00
## `PCIAT-PCIAT_08...10`1                   -1.324774e-01 -2.190753e-01
## `PCIAT-PCIAT_08...10`2                    1.032406e-01 -3.806837e-01
## `PCIAT-PCIAT_08...10`3                    3.232952e-01 -3.133001e-01
## `PCIAT-PCIAT_08...10`4                    5.862834e-01 -3.035017e-02
## `PCIAT-PCIAT_08...10`5                    1.105953e+00  2.377143e+00
## `PCIAT-PCIAT_08...10`Missing             -1.379264e+00 -3.171147e-01
## `PCIAT-PCIAT_09...11`1                    9.227164e-02 -2.296184e-01
## `PCIAT-PCIAT_09...11`2                    1.926308e-01 -3.774076e-01
## `PCIAT-PCIAT_09...11`3                    4.038359e-01 -4.526614e-02
## `PCIAT-PCIAT_09...11`4                    5.271364e-01 -1.710417e-01
## `PCIAT-PCIAT_09...11`5                    7.457430e-01  1.117802e+00
## `PCIAT-PCIAT_09...11`Missing             -1.499005e-01 -1.092401e+00
## `PCIAT-PCIAT_10...12`1                   -2.745534e-01 -1.089779e-01
## `PCIAT-PCIAT_10...12`2                    4.092254e-02 -3.038187e-01
## `PCIAT-PCIAT_10...12`3                    4.273490e-01 -4.646658e-01
## `PCIAT-PCIAT_10...12`4                    5.670661e-01  2.167275e-01
## `PCIAT-PCIAT_10...12`5                    1.078456e+00  1.999181e+00
```

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## `PCIAT-PCIAT_10...12`Missing            -3.267087e+00 -1.910547e+00
## `PCIAT-PCIAT_11...13`1                  -3.261645e-01  2.331157e-01
## `PCIAT-PCIAT_11...13`2                  -1.718260e-01  8.240412e-02
## `PCIAT-PCIAT_11...13`3                   5.323419e-02 -7.759936e-02
## `PCIAT-PCIAT_11...13`4                   4.567903e-01 -1.125908e-01
## `PCIAT-PCIAT_11...13`5                   6.126469e-01 -5.641588e-01
## `PCIAT-PCIAT_11...13`Missing            -1.136285e+00 -1.621600e+00
## `PCIAT-PCIAT_12...14`1                   4.054618e-01 -1.701081e-01
## `PCIAT-PCIAT_12...14`2                   4.619902e-01  9.308766e-02
## `PCIAT-PCIAT_12...14`3                   4.809258e-01  5.516106e-01
## `PCIAT-PCIAT_12...14`4                   1.253212e+00  4.037035e+00
## `PCIAT-PCIAT_12...14`5                  -1.698116e+00 -1.905857e+00
## `PCIAT-PCIAT_12...14`Missing            -3.921825e-01 -1.404961e+00
## `PCIAT-PCIAT_13...15`1                  -1.830707e-01 -2.249164e-01
## `PCIAT-PCIAT_13...15`2                  -2.320238e-02 -1.524079e-01
## `PCIAT-PCIAT_13...15`3                   2.552311e-01 -3.308226e-01
## `PCIAT-PCIAT_13...15`4                   6.211120e-01  5.084063e-02
## `PCIAT-PCIAT_13...15`5                   7.565613e-01  5.405995e-01
## `PCIAT-PCIAT_13...15`Missing            -8.351613e-02 -1.892304e-01
## `PCIAT-PCIAT_14...16`1                   9.327719e-02 -2.567155e-01
## `PCIAT-PCIAT_14...16`2                   2.170889e-01 -3.823109e-01
## `PCIAT-PCIAT_14...16`3                   3.258390e-01 -4.976211e-01
## `PCIAT-PCIAT_14...16`4                   6.322779e-01  2.506618e-01
## `PCIAT-PCIAT_14...16`5                   1.290464e+00  2.222803e+00
## `PCIAT-PCIAT_14...16`Missing             6.458381e-01  1.911301e+00
## `PCIAT-PCIAT_15...17`1                  -3.827085e-01  1.327034e-01
## `PCIAT-PCIAT_15...17`2                  -1.096093e-01 -9.814863e-02
## `PCIAT-PCIAT_15...17`3                   2.414928e-01 -3.270121e-01
## `PCIAT-PCIAT_15...17`4                   4.879278e-01 -2.062636e-01
## `PCIAT-PCIAT_15...17`5                   6.747498e-01  4.728855e-01
## `PCIAT-PCIAT_15...17`Missing            -7.755037e-01 -4.334255e-01
## `PCIAT-PCIAT_16...18`1                  -8.379984e-02  5.250304e-02
## `PCIAT-PCIAT_16...18`2                  -2.767187e-02 -3.612430e-03
## `PCIAT-PCIAT_16...18`3                   1.599884e-01 -2.255822e-02
## `PCIAT-PCIAT_16...18`4                   4.230700e-01 -2.145696e-01
## `PCIAT-PCIAT_16...18`5                   8.008178e-01 -7.144536e-02
## `PCIAT-PCIAT_16...18`Missing            -1.005955e+00 -4.441486e-01
## `PCIAT-PCIAT_17...19`1                  -4.710541e-01  3.538714e-02
## `PCIAT-PCIAT_17...19`2                  -2.719154e-01 -1.932661e-01
## `PCIAT-PCIAT_17...19`3                  -3.843086e-02 -2.871349e-01
## `PCIAT-PCIAT_17...19`4                   4.901393e-01 -9.445213e-02
## `PCIAT-PCIAT_17...19`5                   1.052852e+00  8.395040e-01
## `PCIAT-PCIAT_17...19`Missing            -6.228677e-01  1.172370e+00
## `PCIAT-PCIAT_18`1                       -6.563776e-01  3.685783e-01
## `PCIAT-PCIAT_18`2                       -3.593347e-01  1.326028e-01
## `PCIAT-PCIAT_18`3                       -6.964881e-02  1.770444e-01
## `PCIAT-PCIAT_18`4                        3.348551e-01  2.425145e-02
## `PCIAT-PCIAT_18`5                        9.144616e-01  9.559548e-01
## `PCIAT-PCIAT_18`Missing                 -1.351498e+00 -1.063164e+00
## `PCIAT-PCIAT_19...21`1                   3.463829e-02 -1.026290e-01
## `PCIAT-PCIAT_19...21`2                   1.970456e-01 -2.764248e-01
## `PCIAT-PCIAT_19...21`3                   4.882373e-01  1.110890e-01
## `PCIAT-PCIAT_19...21`4                   6.337856e-01  1.462677e-01
## `PCIAT-PCIAT_19...21`5                   4.555752e-01 -1.127095e-01
```

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## `PCIAT-PCIAT_19...21`Missing          1.308392e-02  1.579375e-01
## `PCIAT-PCIAT_20...22`1                1.428202e-01 -4.157126e-01
## `PCIAT-PCIAT_20...22`2                2.976100e-01 -6.354656e-01
## `PCIAT-PCIAT_20...22`3                4.274457e-01 -4.220023e-01
## `PCIAT-PCIAT_20...22`4                6.128836e-01  3.539504e-01
## `PCIAT-PCIAT_20...22`5                1.173386e+00  3.386336e+00
## `PCIAT-PCIAT_20...22`Missing          2.997430e+00  2.387141e+00
## `PCIAT-PCIAT_Total`1                  1.206091e+00 -4.688591e-01
## `PCIAT-PCIAT_Total`10                 3.147645e+00 -9.470900e-01
## `PCIAT-PCIAT_Total`11                 2.350882e+00 -5.710970e-01
## `PCIAT-PCIAT_Total`12                 2.841291e+00 -8.113774e-01
## `PCIAT-PCIAT_Total`13                 2.550346e+00 -5.684108e-01
## `PCIAT-PCIAT_Total`14                 2.952208e+00 -6.151654e-01
## `PCIAT-PCIAT_Total`15                 2.538536e+00 -6.545566e-01
## `PCIAT-PCIAT_Total`16                 2.698696e+00 -5.211526e-01
## `PCIAT-PCIAT_Total`17                 2.270603e+00 -3.167679e-01
## `PCIAT-PCIAT_Total`18                 2.556298e+00 -2.793730e-01
## `PCIAT-PCIAT_Total`19                 2.050158e+00 -1.462288e-01
## `PCIAT-PCIAT_Total`2                  1.444467e+00 -5.265307e-01
## `PCIAT-PCIAT_Total`20                 1.955178e+00 -1.313884e-01
## `PCIAT-PCIAT_Total`21                 1.843580e+00 -2.895881e-03
## `PCIAT-PCIAT_Total`22                 1.790167e+00  5.514478e-02
## `PCIAT-PCIAT_Total`23                 1.558312e+00  2.516275e-01
## `PCIAT-PCIAT_Total`24                 9.744657e-01  3.537502e-01
## `PCIAT-PCIAT_Total`25                 7.705478e-01  2.358454e-01
## `PCIAT-PCIAT_Total`26                 5.853836e-01  5.599210e-01
## `PCIAT-PCIAT_Total`27                 3.031727e-01  9.730057e-01
## `PCIAT-PCIAT_Total`28                -9.096393e-02  1.103921e+00
## `PCIAT-PCIAT_Total`29                -4.923195e-01  1.052873e+00
## `PCIAT-PCIAT_Total`3                  2.001055e+00 -7.004719e-01
## `PCIAT-PCIAT_Total`30                -6.318668e-01  1.160311e+00
## `PCIAT-PCIAT_Total`31                 3.239635e+00 -9.235472e-01
## `PCIAT-PCIAT_Total`32                 2.655281e+00 -6.812263e-01
## `PCIAT-PCIAT_Total`33                 2.346799e+00 -3.311622e-01
## `PCIAT-PCIAT_Total`34                 2.101939e+00 -3.830857e-01
## `PCIAT-PCIAT_Total`35                 1.331571e+00 -3.897995e-01
## `PCIAT-PCIAT_Total`36                 1.060750e+00 -1.717968e-01
## `PCIAT-PCIAT_Total`37                 9.531515e-01  6.788022e-02
## `PCIAT-PCIAT_Total`38                 7.132085e-01  2.889364e-01
## `PCIAT-PCIAT_Total`39                -2.075743e-01  4.207266e-01
## `PCIAT-PCIAT_Total`4                  2.188227e+00 -8.501591e-01
## `PCIAT-PCIAT_Total`40                -3.657688e-01  2.554192e-01
## `PCIAT-PCIAT_Total`41                -8.979510e-01  5.476162e-01
## `PCIAT-PCIAT_Total`42                -1.247859e+00  2.985439e-01
## `PCIAT-PCIAT_Total`43                -1.945872e+00 -1.822576e-01
## `PCIAT-PCIAT_Total`44                -2.361780e+00  2.657116e-02
## `PCIAT-PCIAT_Total`45                -2.516178e+00  7.861911e-01
## `PCIAT-PCIAT_Total`46                -2.774926e+00  4.916845e-01
## `PCIAT-PCIAT_Total`47                -3.341422e+00  8.577468e-01
## `PCIAT-PCIAT_Total`48                -3.803840e+00  4.235810e-01
## `PCIAT-PCIAT_Total`49                -4.242136e+00  4.961825e-01
## `PCIAT-PCIAT_Total`5                  3.216616e+00 -1.118713e+00
## `PCIAT-PCIAT_Total`50                 5.199126e+00  2.306262e+00
## `PCIAT-PCIAT_Total`51                 4.337958e+00  1.490694e+00
```

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## `PCIAT-PCIAT_Total`52                    4.831500e+00  2.009317e+00
## `PCIAT-PCIAT_Total`53                    3.900004e+00  1.969043e+00
## `PCIAT-PCIAT_Total`54                    3.182698e+00  1.185130e+00
## `PCIAT-PCIAT_Total`55                    2.767027e+00  1.559643e+00
## `PCIAT-PCIAT_Total`56                    2.283904e+00  9.799107e-01
## `PCIAT-PCIAT_Total`57                    1.818144e+00  7.069374e-01
## `PCIAT-PCIAT_Total`58                    1.595316e+00  1.159885e+00
## `PCIAT-PCIAT_Total`59                    1.101853e+00  8.442078e-01
## `PCIAT-PCIAT_Total`6                     2.728877e+00 -1.034662e+00
## `PCIAT-PCIAT_Total`60                    2.463440e-01  4.871080e-01
## `PCIAT-PCIAT_Total`61                   -6.174777e-02  1.058267e+00
## `PCIAT-PCIAT_Total`62                   -3.699166e-01  1.160522e+00
## `PCIAT-PCIAT_Total`63                   -1.151143e+00  2.624664e-01
## `PCIAT-PCIAT_Total`64                   -1.326968e+00 -3.630675e-02
## `PCIAT-PCIAT_Total`65                   -1.719511e+00  4.122092e-01
## `PCIAT-PCIAT_Total`66                   -2.374593e+00 -1.392535e+00
## `PCIAT-PCIAT_Total`67                   -3.380684e+00 -3.014071e+00
## `PCIAT-PCIAT_Total`68                   -3.473552e+00 -6.605630e-01
## `PCIAT-PCIAT_Total`69                   -4.007638e+00 -1.259604e+00
## `PCIAT-PCIAT_Total`7                     2.892512e+00 -9.690122e-01
## `PCIAT-PCIAT_Total`70                   -4.367784e+00 -1.708177e+00
## `PCIAT-PCIAT_Total`71                   -5.444352e+00 -2.704801e+00
## `PCIAT-PCIAT_Total`72                   -5.513516e+00 -2.177559e+00
## `PCIAT-PCIAT_Total`73                   -5.890492e+00 -3.248880e+00
## `PCIAT-PCIAT_Total`74                   -6.628995e+00 -5.930348e+00
## `PCIAT-PCIAT_Total`75                   -7.232010e+00 -6.609600e+00
## `PCIAT-PCIAT_Total`76                   -7.685524e+00 -5.002668e+00
## `PCIAT-PCIAT_Total`77                   -7.682943e+00 -2.854148e+00
## `PCIAT-PCIAT_Total`78                   -8.870154e+00 -7.096168e+00
## `PCIAT-PCIAT_Total`8                     3.239580e+00 -1.178243e+00
## `PCIAT-PCIAT_Total`80                    2.759256e+00  3.820595e+00
## `PCIAT-PCIAT_Total`81                    2.604862e+00  3.969339e+00
## `PCIAT-PCIAT_Total`82                   -2.087032e-01 -4.945356e+00
## `PCIAT-PCIAT_Total`83                    1.402702e+00  2.200346e+00
## `PCIAT-PCIAT_Total`85                   -2.289031e-01  8.206195e-02
## `PCIAT-PCIAT_Total`87                   -1.607076e+00 -3.940845e+00
## `PCIAT-PCIAT_Total`88                   -1.236069e+00 -8.631218e-02
## `PCIAT-PCIAT_Total`89                   -2.000472e+00 -4.130302e+00
## `PCIAT-PCIAT_Total`9                     3.737912e+00 -1.306541e+00
## `PCIAT-PCIAT_Total`90                   -2.579548e+00 -4.826729e+00
## `PCIAT-PCIAT_Total`91                   -2.706017e+00 -3.501876e+00
## `PCIAT-PCIAT_Total`92                   -3.013177e+00 -2.972461e+00
## `PCIAT-PCIAT_Total`93                   -1.698116e+00 -1.905857e+00
## `Basic_Demos-Age`                       -3.648436e-02 -6.265973e-03
## `Basic_Demos-Sex`                       -3.673132e-02 -3.420869e-02
## `CGAS-CGAS_Score`                        1.166706e-03  6.654392e-05
## `Physical-BMI`                          -3.012029e-03 -6.984937e-03
## `Physical-Waist_Circumference`           1.234989e-03 -2.137658e-02
## `Physical-Diastolic_BP`                 -9.296587e-04  9.098043e-05
## `Physical-HeartRate`                    -3.781786e-04 -2.110055e-03
## `Physical-Systolic_BP`                  -1.944532e-04 -3.463856e-05
## `Fitness_Endurance-Max_Stage`           -8.295668e-03  5.536731e-02
## `Fitness_Endurance-Time_Mins`           -1.012037e-02 -1.607397e-02
## `Fitness_Endurance-Time_Sec`             1.539511e-03 -1.827601e-03
```

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## `FGC-FGC_CU`                              4.039038e-03 -3.957323e-03
## `FGC-FGC_CU_Zone`                        -5.374633e-02  9.950935e-02
## `FGC-FGC_GSND_Zone`                      -1.583229e-01  7.329932e-02
## `FGC-FGC_GSD`                            -2.916233e-03 -1.271059e-02
## `FGC-FGC_GSD_Zone`                        1.800446e-01 -4.683080e-02
## `FGC-FGC_PU`                              5.581678e-04  1.366593e-02
## `FGC-FGC_PU_Zone`                         7.409414e-02 -7.963237e-02
## `FGC-FGC_SRL`                             8.176816e-03  7.339817e-03
## `FGC-FGC_SRL_Zone`                       -8.431032e-02 -4.465430e-02
## `FGC-FGC_SRR_Zone`                        8.585353e-02 -1.067110e-01
## `FGC-FGC_TL`                             -1.302130e-02 -2.798136e-02
## `FGC-FGC_TL_Zone`                         6.651262e-02  1.472454e-01
## `BIA-BIA_Activity_Level_num`              2.015970e-03  5.518076e-02
## `BIA-BIA_BMI`                             4.190911e-01  6.570282e-01
## `BIA-BIA_BMR`                             3.045479e-05  3.871908e-06
## `BIA-BIA_FFMI`                           -4.110243e-01 -6.347314e-01
## `BIA-BIA_FMI`                            -4.087832e-01 -6.174119e-01
## `BIA-BIA_Frame_num`                      -8.668160e-02 -1.686427e-01
## `PAQ_A-PAQ_A_Total`                       8.411569e-02  1.071958e-01
## `PAQ_C-PAQ_C_Total`                       3.709851e-02 -2.865201e-02
## `PCIAT-PCIAT_01...55`                     1.944691e-02  2.453051e-02
## `PCIAT-PCIAT_02...56`                     1.921404e-01 -8.805520e-02
## `PCIAT-PCIAT_03...57`                    -4.377950e-01 -1.393485e-02
## `PCIAT-PCIAT_04...58`                     2.995870e-01  7.270028e-02
## `PCIAT-PCIAT_05...59`                     8.776840e-03 -1.019906e-01
## `PCIAT-PCIAT_06...60`                     3.411999e-01  9.437103e-02
## `PCIAT-PCIAT_07...61`                     2.586162e-01  6.391938e-02
## `PCIAT-PCIAT_08...62`                     2.942659e-01 -2.674696e-02
## `PCIAT-PCIAT_09...63`                     3.035618e-01 -3.538968e-02
## `PCIAT-PCIAT_10...64`                     2.641400e-01  3.934108e-03
## `PCIAT-PCIAT_11...65`                     1.250738e-01 -7.230403e-02
## `PCIAT-PCIAT_12...66`                     3.667496e-01  9.383383e-02
## `PCIAT-PCIAT_13...67`                     2.509405e-01 -7.208216e-02
## `PCIAT-PCIAT_14...68`                     3.663938e-01  2.367516e-02
## `PCIAT-PCIAT_15...69`                     2.059477e-01 -6.577247e-02
## `PCIAT-PCIAT_16...70`                     2.326189e-01 -5.421167e-02
## `PCIAT-PCIAT_17...71`                     9.201862e-02 -4.497138e-02
## `PCIAT-PCIAT_19...72`                     3.154230e-01 -5.791657e-02
## `PCIAT-PCIAT_20...73`                     4.213444e-01  1.380740e-02
## `SDS-SDS_Total_T`                        -4.366353e-03 -2.039933e-03
## `PreInt_EduHx-computerinternet_hoursday` -9.901346e-03  6.305781e-03
##                                                     LD3
## `PCIAT-PCIAT_01...3`1                    -1.586845e-01
## `PCIAT-PCIAT_01...3`2                    -3.094578e-01
## `PCIAT-PCIAT_01...3`3                    -1.258065e-01
## `PCIAT-PCIAT_01...3`4                     1.189588e-01
## `PCIAT-PCIAT_01...3`5                     2.166484e-02
## `PCIAT-PCIAT_01...3`Missing               3.666565e+00
## `PCIAT-PCIAT_02...4`1                    -9.634444e-02
## `PCIAT-PCIAT_02...4`2                    -5.994161e-02
## `PCIAT-PCIAT_02...4`3                     4.344406e-01
## `PCIAT-PCIAT_02...4`4                    -1.984205e-02
## `PCIAT-PCIAT_02...4`5                    -3.721594e-02
## `PCIAT-PCIAT_02...4`Missing               1.572208e+00
```

```
## `PCIAT-PCIAT_03...5`1                    -3.795493e-01
## `PCIAT-PCIAT_03...5`2                    -2.564305e-01
## `PCIAT-PCIAT_03...5`3                     2.689468e-01
## `PCIAT-PCIAT_03...5`4                     1.698413e-01
## `PCIAT-PCIAT_03...5`5                    -2.211617e-01
## `PCIAT-PCIAT_03...5`Missing               1.322173e+00
## `PCIAT-PCIAT_04...6`1                     2.212145e-02
## `PCIAT-PCIAT_04...6`2                    -1.183302e-01
## `PCIAT-PCIAT_04...6`3                    -1.052367e-01
## `PCIAT-PCIAT_04...6`4                    -7.351236e-01
## `PCIAT-PCIAT_04...6`5                     1.697848e+00
## `PCIAT-PCIAT_04...6`Missing              -1.785909e+00
## `PCIAT-PCIAT_05...7`1                    -4.224914e-01
## `PCIAT-PCIAT_05...7`2                    -5.376312e-02
## `PCIAT-PCIAT_05...7`3                     3.697158e-01
## `PCIAT-PCIAT_05...7`4                     3.718103e-01
## `PCIAT-PCIAT_05...7`5                    -2.222260e-01
## `PCIAT-PCIAT_05...7`Missing               1.470415e+00
## `PCIAT-PCIAT_06...8`1                     1.004754e-01
## `PCIAT-PCIAT_06...8`2                     5.596365e-02
## `PCIAT-PCIAT_06...8`3                    -1.690507e-01
## `PCIAT-PCIAT_06...8`4                    -6.365638e-01
## `PCIAT-PCIAT_06...8`5                     5.383043e-01
## `PCIAT-PCIAT_06...8`Missing              -5.693373e-02
## `PCIAT-PCIAT_07...9`1                     3.968132e-02
## `PCIAT-PCIAT_07...9`2                     2.589811e-01
## `PCIAT-PCIAT_07...9`3                    -1.351013e-01
## `PCIAT-PCIAT_07...9`4                    -1.934777e-01
## `PCIAT-PCIAT_07...9`5                     7.094449e-01
## `PCIAT-PCIAT_07...9`Missing              -7.501432e-01
## `PCIAT-PCIAT_08...10`1                    9.595258e-02
## `PCIAT-PCIAT_08...10`2                   -1.986115e-03
## `PCIAT-PCIAT_08...10`3                   -3.313020e-01
## `PCIAT-PCIAT_08...10`4                   -6.212189e-01
## `PCIAT-PCIAT_08...10`5                    1.822913e+00
## `PCIAT-PCIAT_08...10`Missing              1.114026e-01
## `PCIAT-PCIAT_09...11`1                    2.289925e-01
## `PCIAT-PCIAT_09...11`2                    2.541799e-01
## `PCIAT-PCIAT_09...11`3                   -3.272961e-01
## `PCIAT-PCIAT_09...11`4                   -8.103760e-01
## `PCIAT-PCIAT_09...11`5                    3.008627e-01
## `PCIAT-PCIAT_09...11`Missing              2.739988e+00
## `PCIAT-PCIAT_10...12`1                    1.961242e-01
## `PCIAT-PCIAT_10...12`2                    2.586390e-01
## `PCIAT-PCIAT_10...12`3                   -1.926696e-01
## `PCIAT-PCIAT_10...12`4                   -7.762389e-01
## `PCIAT-PCIAT_10...12`5                    1.197525e+00
## `PCIAT-PCIAT_10...12`Missing             -8.328341e+00
## `PCIAT-PCIAT_11...13`1                   -1.796741e-01
## `PCIAT-PCIAT_11...13`2                    1.665676e-02
## `PCIAT-PCIAT_11...13`3                    1.097826e-01
## `PCIAT-PCIAT_11...13`4                    1.195964e-01
## `PCIAT-PCIAT_11...13`5                   -4.719430e-01
## `PCIAT-PCIAT_11...13`Missing             -3.284290e+00
```

```
## `PCIAT-PCIAT_12...14`1                  -3.497841e-02
## `PCIAT-PCIAT_12...14`2                   2.653811e-01
## `PCIAT-PCIAT_12...14`3                   1.031616e+00
## `PCIAT-PCIAT_12...14`4                   1.591990e-01
## `PCIAT-PCIAT_12...14`5                  -9.516057e-01
## `PCIAT-PCIAT_12...14`Missing            -2.656500e+00
## `PCIAT-PCIAT_13...15`1                   2.299469e-01
## `PCIAT-PCIAT_13...15`2                   2.456798e-01
## `PCIAT-PCIAT_13...15`3                   2.505244e-02
## `PCIAT-PCIAT_13...15`4                  -3.189470e-01
## `PCIAT-PCIAT_13...15`5                  -5.267410e-02
## `PCIAT-PCIAT_13...15`Missing            -2.688380e-01
## `PCIAT-PCIAT_14...16`1                   2.159530e-01
## `PCIAT-PCIAT_14...16`2                   6.436816e-02
## `PCIAT-PCIAT_14...16`3                  -1.151560e-01
## `PCIAT-PCIAT_14...16`4                  -8.690420e-01
## `PCIAT-PCIAT_14...16`5                   1.004166e+00
## `PCIAT-PCIAT_14...16`Missing             4.310515e-01
## `PCIAT-PCIAT_15...17`1                   1.110495e-04
## `PCIAT-PCIAT_15...17`2                   3.431932e-01
## `PCIAT-PCIAT_15...17`3                   8.514713e-02
## `PCIAT-PCIAT_15...17`4                  -2.699936e-01
## `PCIAT-PCIAT_15...17`5                  -1.068208e-02
## `PCIAT-PCIAT_15...17`Missing            -2.098289e+00
## `PCIAT-PCIAT_16...18`1                   1.673270e-01
## `PCIAT-PCIAT_16...18`2                   1.971226e-01
## `PCIAT-PCIAT_16...18`3                   4.443918e-01
## `PCIAT-PCIAT_16...18`4                  -2.472913e-01
## `PCIAT-PCIAT_16...18`5                  -4.779412e-01
## `PCIAT-PCIAT_16...18`Missing             2.248667e-01
## `PCIAT-PCIAT_17...19`1                   3.520385e-02
## `PCIAT-PCIAT_17...19`2                   4.126824e-01
## `PCIAT-PCIAT_17...19`3                   3.325225e-01
## `PCIAT-PCIAT_17...19`4                  -5.098104e-01
## `PCIAT-PCIAT_17...19`5                  -5.228566e-01
## `PCIAT-PCIAT_17...19`Missing             6.860828e-01
## `PCIAT-PCIAT_18`1                       -3.938983e-01
## `PCIAT-PCIAT_18`2                       -2.350474e-01
## `PCIAT-PCIAT_18`3                       -5.435822e-01
## `PCIAT-PCIAT_18`4                       -1.182895e+00
## `PCIAT-PCIAT_18`5                       -9.998151e-01
## `PCIAT-PCIAT_18`Missing                  2.935671e-01
## `PCIAT-PCIAT_19...21`1                  -2.035193e-04
## `PCIAT-PCIAT_19...21`2                   1.075498e-01
## `PCIAT-PCIAT_19...21`3                  -2.238243e-02
## `PCIAT-PCIAT_19...21`4                  -3.688956e-01
## `PCIAT-PCIAT_19...21`5                   3.855386e-01
## `PCIAT-PCIAT_19...21`Missing            -2.851416e-02
## `PCIAT-PCIAT_20...22`1                   1.688891e-01
## `PCIAT-PCIAT_20...22`2                  -1.380674e-01
## `PCIAT-PCIAT_20...22`3                  -6.170397e-01
## `PCIAT-PCIAT_20...22`4                  -6.034937e-01
## `PCIAT-PCIAT_20...22`5                   2.828689e+00
## `PCIAT-PCIAT_20...22`Missing             6.668921e+00
```

```
## `PCIAT-PCIAT_Total`1                    3.233155e-01
## `PCIAT-PCIAT_Total`10                   5.768523e-01
## `PCIAT-PCIAT_Total`11                   1.608214e-01
## `PCIAT-PCIAT_Total`12                   3.490776e-01
## `PCIAT-PCIAT_Total`13                   2.511662e-01
## `PCIAT-PCIAT_Total`14                   4.101953e-01
## `PCIAT-PCIAT_Total`15                   4.842111e-01
## `PCIAT-PCIAT_Total`16                   2.455045e-01
## `PCIAT-PCIAT_Total`17                   1.847240e-01
## `PCIAT-PCIAT_Total`18                   1.139747e-01
## `PCIAT-PCIAT_Total`19                   9.397432e-02
## `PCIAT-PCIAT_Total`2                    3.421280e-01
## `PCIAT-PCIAT_Total`20                   9.111841e-02
## `PCIAT-PCIAT_Total`21                  -8.225023e-02
## `PCIAT-PCIAT_Total`22                  -7.408968e-02
## `PCIAT-PCIAT_Total`23                  -2.165987e-01
## `PCIAT-PCIAT_Total`24                  -3.745243e-01
## `PCIAT-PCIAT_Total`25                  -3.084484e-01
## `PCIAT-PCIAT_Total`26                  -5.991158e-01
## `PCIAT-PCIAT_Total`27                  -6.387841e-01
## `PCIAT-PCIAT_Total`28                  -7.765314e-01
## `PCIAT-PCIAT_Total`29                  -6.961618e-01
## `PCIAT-PCIAT_Total`3                    4.617114e-01
## `PCIAT-PCIAT_Total`30                  -1.032439e+00
## `PCIAT-PCIAT_Total`31                  -2.597515e-01
## `PCIAT-PCIAT_Total`32                  -3.005259e-01
## `PCIAT-PCIAT_Total`33                  -4.994080e-01
## `PCIAT-PCIAT_Total`34                  -5.225019e-01
## `PCIAT-PCIAT_Total`35                  -2.215847e-01
## `PCIAT-PCIAT_Total`36                  -2.354348e-01
## `PCIAT-PCIAT_Total`37                  -5.154645e-01
## `PCIAT-PCIAT_Total`38                  -3.132745e-01
## `PCIAT-PCIAT_Total`39                  -2.596933e-01
## `PCIAT-PCIAT_Total`4                    5.551025e-01
## `PCIAT-PCIAT_Total`40                  -1.035392e-01
## `PCIAT-PCIAT_Total`41                  -1.962813e-01
## `PCIAT-PCIAT_Total`42                   2.374782e-01
## `PCIAT-PCIAT_Total`43                   8.394449e-01
## `PCIAT-PCIAT_Total`44                   5.247435e-01
## `PCIAT-PCIAT_Total`45                   4.384591e-01
## `PCIAT-PCIAT_Total`46                   1.360433e-01
## `PCIAT-PCIAT_Total`47                   4.300780e-01
## `PCIAT-PCIAT_Total`48                   1.304094e+00
## `PCIAT-PCIAT_Total`49                   1.249912e+00
## `PCIAT-PCIAT_Total`5                    7.559454e-01
## `PCIAT-PCIAT_Total`50                  -1.095035e+00
## `PCIAT-PCIAT_Total`51                  -1.051202e+00
## `PCIAT-PCIAT_Total`52                  -8.898581e-01
## `PCIAT-PCIAT_Total`53                  -6.613414e-01
## `PCIAT-PCIAT_Total`54                  -9.515009e-01
## `PCIAT-PCIAT_Total`55                  -6.063124e-01
## `PCIAT-PCIAT_Total`56                  -1.053335e+00
## `PCIAT-PCIAT_Total`57                  -1.001262e+00
## `PCIAT-PCIAT_Total`58                   1.517157e-01
```

```
## `PCIAT-PCIAT_Total`59                    -6.350156e-01
## `PCIAT-PCIAT_Total`6                      7.434342e-01
## `PCIAT-PCIAT_Total`60                     4.425836e-01
## `PCIAT-PCIAT_Total`61                     1.281713e+00
## `PCIAT-PCIAT_Total`62                     4.684912e-01
## `PCIAT-PCIAT_Total`63                     1.521951e-01
## `PCIAT-PCIAT_Total`64                     6.681073e-01
## `PCIAT-PCIAT_Total`65                     1.495293e+00
## `PCIAT-PCIAT_Total`66                     5.384027e-01
## `PCIAT-PCIAT_Total`67                    -1.077272e-01
## `PCIAT-PCIAT_Total`68                     2.331815e+00
## `PCIAT-PCIAT_Total`69                     1.553832e+00
## `PCIAT-PCIAT_Total`7                      6.507698e-01
## `PCIAT-PCIAT_Total`70                     9.901237e-01
## `PCIAT-PCIAT_Total`71                     2.843302e+00
## `PCIAT-PCIAT_Total`72                     2.134115e+00
## `PCIAT-PCIAT_Total`73                    -2.497927e-02
## `PCIAT-PCIAT_Total`74                    -1.047310e+00
## `PCIAT-PCIAT_Total`75                    -1.148931e+00
## `PCIAT-PCIAT_Total`76                     7.629510e-01
## `PCIAT-PCIAT_Total`77                     2.506949e+00
## `PCIAT-PCIAT_Total`78                    -2.047580e+00
## `PCIAT-PCIAT_Total`8                      7.879867e-01
## `PCIAT-PCIAT_Total`80                     3.221014e+00
## `PCIAT-PCIAT_Total`81                     2.897619e+00
## `PCIAT-PCIAT_Total`82                    -7.032204e+00
## `PCIAT-PCIAT_Total`83                     1.507237e+00
## `PCIAT-PCIAT_Total`85                     1.209811e+00
## `PCIAT-PCIAT_Total`87                    -3.415681e+00
## `PCIAT-PCIAT_Total`88                    -3.288431e-01
## `PCIAT-PCIAT_Total`89                    -6.197886e+00
## `PCIAT-PCIAT_Total`9                      8.972877e-01
## `PCIAT-PCIAT_Total`90                    -2.829074e+00
## `PCIAT-PCIAT_Total`91                     3.684787e+00
## `PCIAT-PCIAT_Total`92                    -2.932079e+00
## `PCIAT-PCIAT_Total`93                    -9.516057e-01
## `Basic_Demos-Age`                         8.463502e-03
## `Basic_Demos-Sex`                        -1.742665e-01
## `CGAS-CGAS_Score`                        -6.589716e-03
## `Physical-BMI`                            1.636936e-02
## `Physical-Waist_Circumference`           -1.114040e-02
## `Physical-Diastolic_BP`                  -2.562546e-03
## `Physical-HeartRate`                     -6.778800e-04
## `Physical-Systolic_BP`                    6.113774e-03
## `Fitness_Endurance-Max_Stage`            -9.704406e-02
## `Fitness_Endurance-Time_Mins`             3.973209e-02
## `Fitness_Endurance-Time_Sec`              2.030224e-03
## `FGC-FGC_CU`                              6.503346e-03
## `FGC-FGC_CU_Zone`                        -5.742163e-02
## `FGC-FGC_GSND_Zone`                       7.406683e-02
## `FGC-FGC_GSD`                            -1.762128e-02
## `FGC-FGC_GSD_Zone`                        4.496083e-01
## `FGC-FGC_PU`                              8.004880e-03
## `FGC-FGC_PU_Zone`                        -1.119696e-01
```

```
## `FGC-FGC_SRL`                              1.124144e-02
## `FGC-FGC_SRL_Zone`                        -2.308958e-01
## `FGC-FGC_SRR_Zone`                         8.588055e-02
## `FGC-FGC_TL`                               4.465618e-04
## `FGC-FGC_TL_Zone`                          1.162742e-01
## `BIA-BIA_Activity_Level_num`              -1.501567e-02
## `BIA-BIA_BMI`                             -9.300717e-01
## `BIA-BIA_BMR`                             -3.542760e-06
## `BIA-BIA_FFMI`                             9.034422e-01
## `BIA-BIA_FMI`                              9.231212e-01
## `BIA-BIA_Frame_num`                        1.788803e-02
## `PAQ_A-PAQ_A_Total`                        1.043723e-01
## `PAQ_C-PAQ_C_Total`                       -5.527090e-02
## `PCIAT-PCIAT_01...55`                     -5.197060e-02
## `PCIAT-PCIAT_02...56`                      7.164047e-02
## `PCIAT-PCIAT_03...57`                      9.158956e-03
## `PCIAT-PCIAT_04...58`                     -2.566676e-02
## `PCIAT-PCIAT_05...59`                      1.193839e-01
## `PCIAT-PCIAT_06...60`                     -4.424819e-02
## `PCIAT-PCIAT_07...61`                      5.566956e-02
## `PCIAT-PCIAT_08...62`                     -2.874887e-02
## `PCIAT-PCIAT_09...63`                     -4.140312e-04
## `PCIAT-PCIAT_10...64`                      5.368919e-03
## `PCIAT-PCIAT_11...65`                     -1.904831e-02
## `PCIAT-PCIAT_12...66`                      4.541000e-02
## `PCIAT-PCIAT_13...67`                      4.595264e-02
## `PCIAT-PCIAT_14...68`                     -1.104276e-02
## `PCIAT-PCIAT_15...69`                      4.844843e-02
## `PCIAT-PCIAT_16...70`                      2.351184e-02
## `PCIAT-PCIAT_17...71`                      4.497675e-02
## `PCIAT-PCIAT_19...72`                      1.128741e-03
## `PCIAT-PCIAT_20...73`                      7.224358e-02
## `SDS-SDS_Total_T`                          2.370364e-03
## `PreInt_EduHx-computerinternet_hoursday` -4.792995e-02
##
## Proportion of trace:
##    LD1    LD2    LD3
## 0.9120 0.0715 0.0165
```

```r
# Predicting using the LDA model
lda_predictions_train = predict(lda_model_train, lda_data_train)

# Add predictions to the original data
lda_data_train$predicted_sii = lda_predictions_train$class

# Confusion matrix to evaluate classification accuracy
confusion_matrix_train = table(lda_data_train$sii, lda_data_train$predicted_sii)

print("Train Confusion Matrix:")
```

```
## [1] "Train Confusion Matrix:"
```

```r
print(confusion_matrix_train)
```

```
##
```

```
##        0    1    2    3
##   0 1250   26    0    0
##   1    0  584    0    0
##   2    0    0  303    0
##   3    0    0    0   28
```

```r
# Calculate accuracy
accuracy_train = sum(diag(confusion_matrix_train)) / sum(confusion_matrix_train)
print(paste("Train Accuracy:", round(accuracy_train * 100, 2), "%"))
```

```
## [1] "Train Accuracy: 98.81 %"
```

```r
error_rate_train = 1 - accuracy_train
print(paste("Train Error Rate:", round(error_rate_train * 100, 2), "%"))
```

```
## [1] "Train Error Rate: 1.19 %"
```

```r
# ----------------------------------------------------------------------------------------------------
# Testing on the Test Set

lda_data_test = test_set
lda_data_test$sii = as.factor(lda_data_test$sii)

# Remove 'id' column or any non-predictor columns, if necessary
lda_data_test = subset(lda_data_test, select = -id)

# Perform Linear Discriminant Analysis on the test set
lda_model_test = lda(sii ~ ., data = lda_data_test)
```

```
## Warning in lda.default(x, grouping, ...): variables are collinear
```

```r
# Predict using the LDA model on the test set
lda_predictions_test = predict(lda_model_test, lda_data_test)

# Add predictions to the original data
lda_data_test$predicted_sii = lda_predictions_test$class

# Confusion matrix for the test set
confusion_matrix_test = table(lda_data_test$sii, lda_data_test$predicted_sii)

print("Test Confusion Matrix:")
```

```
## [1] "Test Confusion Matrix:"
```

```r
kable(confusion_matrix_test)
```

	0	1	2	3
0	314	4	0	0
1	0	146	0	0
2	0	0	75	0
3	0	0	0	6

```r
# Calculate accuracy on the test set
accuracy_test = sum(diag(confusion_matrix_test)) / sum(confusion_matrix_test)
print(paste("Test Accuracy:", round(accuracy_test * 100, 2), "%"))
```

```
## [1] "Test Accuracy: 99.27 %"
```

```r
error_rate_test = 1 - accuracy_test
print(paste("Test Error Rate:", round(error_rate_test * 100, 2), "%"))
```

```
## [1] "Test Error Rate: 0.73 %"
```